BOOK 3
DYNAMICS

3 Dynamics

An understanding of forces from Cambridge IGCSE/O Level Physics or equivalent is assumed.

3.1 Momentum and Newton's laws of motion

Candidates should be able to:

1 understand that mass is the property of an object that resists change in motion

2 recall $F = ma$ and solve problems using it, understanding that acceleration and resultant force are always in the same direction

3 define and use linear momentum as the product of mass and velocity

4 define and use force as rate of change of momentum

5 state and apply each of Newton's laws of motion

6 describe and use the concept of weight as the effect of a gravitational field on a mass and recall that the weight of an object is equal to the product of its mass and the acceleration of free fall

3.2 Non-uniform motion

Candidates should be able to:

1 show a qualitative understanding of frictional forces and viscous/drag forces including air resistance (no treatment of the coefficients of friction and viscosity is required, and a simple model of drag force increasing as speed increases is sufficient)

2 describe and explain qualitatively the motion of objects in a uniform gravitational field with air resistance

3 understand that objects moving against a resistive force may reach a terminal (constant) velocity

3.3 Linear momentum and its conservation

Candidates should be able to:

1 state the principle of conservation of momentum

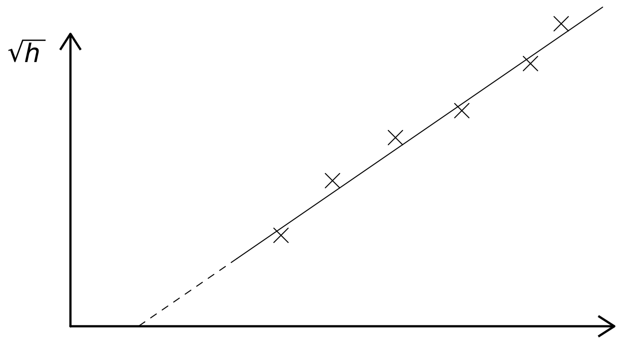
2 apply the principle of conservation of momentum to solve simple problems, including elastic and inelastic interactions between objects in both one and two dimensions (knowledge of the concept of coefficient of restitution is not required)

3 recall that, for a perfectly elastic collision, the relative speed of approach is equal to the relative speed of separation

4 understand that, while momentum of a system is always conserved in interactions between objects, some change in kinetic energy may take place

1 **9702/1/M/J/02/Q4**

A student measures the time t for a ball to fall from rest through a vertical distance h. Knowing that the equation $h = \frac{1}{2} gt^2$ applies, the student plots the graph shown.



Which of the following is an explanation for the intercept on the t axis?

A Air resistance has not been taken into account for larger values of h.

B There is a constant delay between starting the timer and releasing the ball.

C There is an error in the timer that consistently makes it run fast.

D The student should have plotted h against t^2.

2 **9702/11/O/N/10/Q11***

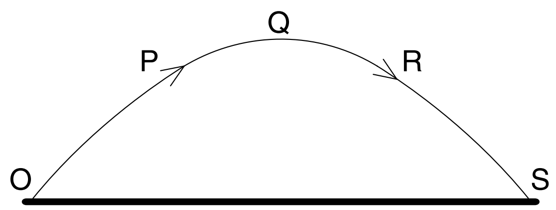
A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\dfrac{v_1}{v_2}$?

A $\dfrac{M_1}{M_2}$ **B** $\dfrac{M_2}{M_1}$ **C** $\left(\dfrac{M_1}{M_2}\right)^{\frac{1}{2}}$ **D** $\left(\dfrac{M_2}{M_1}\right)^{\frac{1}{2}}$

3 **9702/12/M/J/12/Q10**

A projectile is launched at point O and follows the path OPQRS, as shown. Air resistance may be neglected.



Which statement is true for the projectile when it is at the highest point Q of its path?

A The horizontal component of the projectile's acceleration is zero.

B The horizontal component of the projectile's velocity is zero.

C The kinetic energy of the projectile is zero.

D The momentum of the projectile is zero.

4 **9702/01/M/J/07/Q9***

What is meant by the weight of an object?

A the gravitational field acting on the object

B the gravitational force acting on the object

C the mass of the object multiplied by gravity

D the object's mass multiplied by its acceleration

5 **9702/1/O/N/02/Q11**

Two spheres A and B approach each other along the same straight line with speeds u_A and u_B. The spheres collide and move off with speeds v_A and v_B, both in the same direction as the initial direction of sphere A, as shown below.

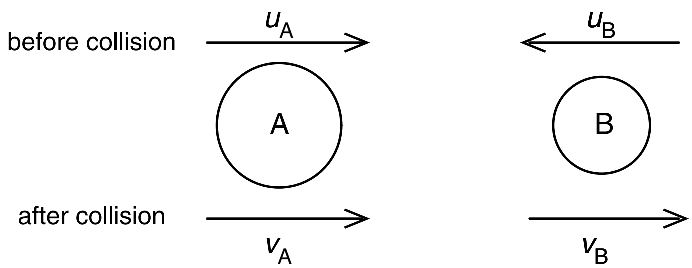
Which equation applies to an elastic collision?

A $u_A + u_B = v_B - v_A$

B $u_A - u_B = v_B - v_A$

C $u_A - u_B = v_B + v_A$

D $u_A + u_B = v_B + v_A$



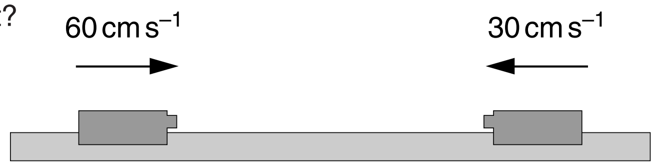
6 **9702/1/O/N/02/Q12**

Two equal masses travel towards each other on a frictionless air track at speeds of $60\,\text{cm}\,\text{s}^{-1}$ and $30\,\text{cm}\,\text{s}^{-1}$. They stick together on impact.

What is the speed of the masses after impact?

A $15\,\text{cm}\,\text{s}^{-1}$ **B** $20\,\text{cm}\,\text{s}^{-1}$

C $30\,\text{cm}\,\text{s}^{-1}$ **D** $45\,\text{cm}\,\text{s}^{-1}$



7 **9702/01/M/J/03/Q12**

A ball of mass 2 kg travelling at $8\,\text{m}\,\text{s}^{-1}$ strikes a ball of mass 4 kg travelling at $2\,\text{m}\,\text{s}^{-1}$. Both balls are moving along the same straight line as shown.

After collision, both balls move at the same velocity v.

What is the magnitude of the velocity v?

A $4\,\text{m}\,\text{s}^{-1}$ **B** $5\,\text{m}\,\text{s}^{-1}$ **C** $6\,\text{m}\,\text{s}^{-1}$ **D** $8\,\text{m}\,\text{s}^{-1}$

8 **9702/01/O/N/03/Q9***

Which of the following is a statement of the principle of conservation of momentum?

A Momentum is the product of mass and velocity.

B In an elastic collision, momentum is constant.

C The momentum of an isolated system is constant.

D The force acting on a body is proportional to its rate of change of momentum.

9 **9702/01/O/N/08/Q9***

A ball falls vertically and bounces on the ground.

The following statements are about the forces acting while the ball is in contact with the ground.

Which statement is correct?

A The force that the ball exerts on the ground is always equal to the weight of the ball.

B The force that the ball exerts on the ground is always equal in magnitude and opposite in direction to the force the ground exerts on the ball.

C The force that the ball exerts on the ground is always less than the weight of the ball.

D The weight of the ball is always equal in magnitude and opposite in direction to the force that the ground exerts on the ball.

10 9702/01/O/N/03/Q11

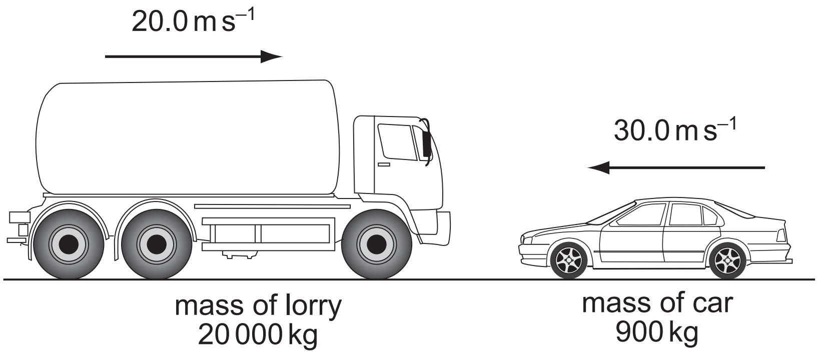
A molecule of mass m travelling horizontally with velocity u hits a vertical wall at right angles to the wall. It then rebounds horizontally with the same speed.

What is its change in momentum?

A zero **B** mu **C** $-mu$ **D** $-2mu$

11 9702/01/M/J/04/Q11

The diagram shows a situation just before a head-on collision. A lorry of mass 20 000 kg is travelling at $20.0\,m\,s^{-1}$ towards a car of mass 900 kg travelling at $30.0\,m\,s^{-1}$ towards the lorry.



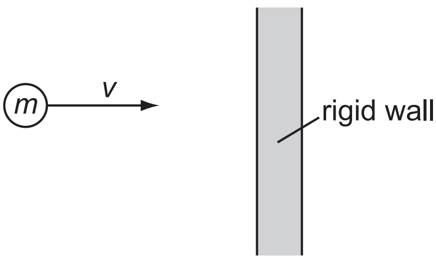
What is the magnitude of the total momentum?

A 373 kN s **B** 427 kN s

C 3600 kN s **D** 4410 kN s

12 9702/01/O/N/04/Q11

A particle of mass m strikes a vertical rigid wall perpendicularly from the left with velocity v.



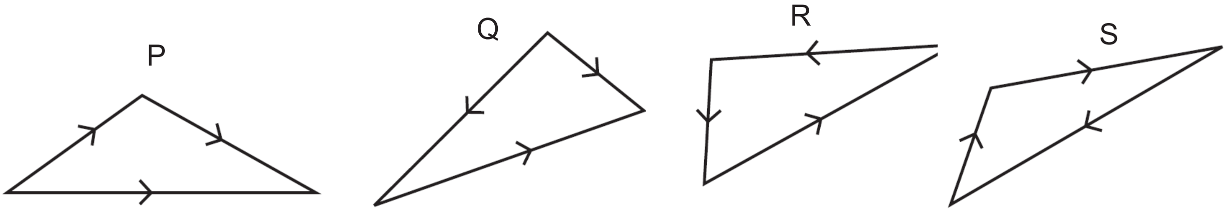
If the collision is perfectly elastic, the total change in momentum of the particle that occurs as a result of the collision is

A $2mv$ to the right. **B** $2mv$ to the left.

C mv to the right. **D** mv to the left.

13 9702/01/O/N/04/Q12

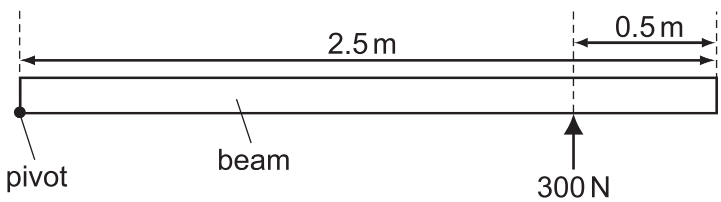
Which two vector diagrams represent forces in equilibrium?



A P and Q **B** Q and R **C** R and S **D** S and P

14 9702/01/O/N/04/Q13

A long uniform beam is pivoted at one end. A force of 300 N is applied to hold the beam horizontally.



What is the weight of the beam?

A 300 N **B** 480 N

C 600 N **D** 960 N

15 **9702/01/M/J/05/Q10**

Which is **not** one of Newton's laws of motion?

A The total momentum of a system of interacting bodies remains constant, providing no external force acts.
B The rate of change of momentum of a body is directly proportional to the external force acting on the body and takes place in the direction of the force.
C If body A exerts a force on body B, then body B exerts an equal and oppositely-directed force on body A.
D A body continues in a state of rest or of uniform motion in a straight line unless acted upon by some external force.

16 **9702/01/M/J/05/Q11***

Two equal masses travel towards each other on a frictionless air track at speeds of 60 cm s^{-1} and 40 cm s^{-1}. They stick together on impact.

What is the speed of the masses after impact?

A 10 cm s^{-1} **B** 20 cm s^{-1}
C 40 cm s^{-1} **D** 50 cm s^{-1}

17 **9702/01/M/J/05/Q12***

What is the centre of gravity of an object?

A the geometrical centre of the object
B the point about which the total torque is zero
C the point at which the weight of the object may be considered to act
D the point through which gravity acts

18 **9702/01/O/N/05/Q9**

Which is a statement of the principle of conservation of momentum?

A A force is equal to the rate of change of momentum of the body upon which it acts.
B In a perfectly elastic collision, the relative momentum of the bodies before impact is equal to their relative momentum after impact.
C The momentum of a body is the product of the mass of the body and its velocity.
D The total momentum of a system of interacting bodies remains constant, providing no external force acts.

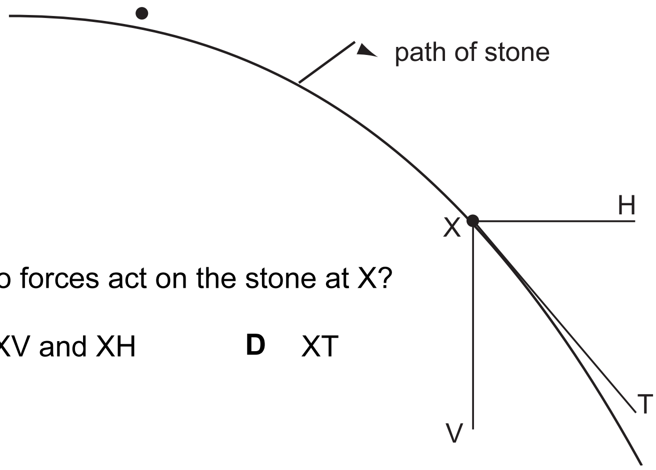
19 **9702/01/O/N/05/Q10**

The gravitational field strength on the surface of planet P is one tenth of that on the surface of planet Q.
On the surface of P, a body has its mass measured to be 1.0 kg and its weight measured to be 1.0 N.
What results are obtained for measurements of the mass and weight of the same body on the surface of planet Q?

	mass on Q	weight on Q
A	1.0 kg	0.1 N
B	1.0 kg	10 N
C	10 kg	10 N
D	10 kg	100 N

20 **9702/01/O/N/05/Q11**

A stone is projected horizontally in a vacuum and moves along a path as shown. X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.



Along which direction or directions do forces act on the stone at X?

A XV **B** XH **C** XV and XH **D** XT

21 **9702/01/M/J/06/Q10**

A cyclist is riding at a steady speed on a level road.

According to Newton's third law of motion, what is equal and opposite to the backward push of the back wheel on the road?

A the force exerted by the cyclist on the pedals

B the forward push of the road on the back wheel

C the tension in the cycle chain

D the total air resistance and friction force

22 **9702/01/M/J/06/Q11**

In perfectly elastic collisions between two atoms, it is always true to say that

A the initial speed of one atom will be the same as the final speed of the other atom.

B the relative speed of approach between the two atoms equals their relative speed of separation.

C the total momentum must be conserved, but a small amount of the total kinetic energy may be lost in the collision.

D whatever their initial states of motion, neither atom can be stationary after the collision.

23 **9702/01/M/J/06/Q12***

Two railway trucks of masses *m* and 3*m* move towards each other in opposite directions with speeds 2*v* and *v* respectively. These trucks collide and stick together.

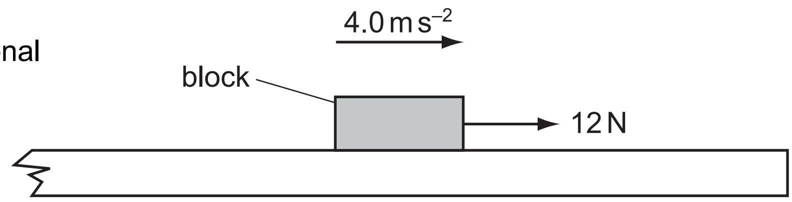
What is the speed of the trucks after the collision?

A $\frac{v}{4}$ **B** $\frac{v}{2}$ **C** v **D** $\frac{5v}{4}$

24 **9702/01/O/N/07/Q10**

A block of mass 0.60 kg is on a rough horizontal surface. A force of 12 N is applied to the block and it accelerates at $4.0\,\mathrm{m\,s^{-2}}$.

What is the magnitude of the frictional force acting on the block?

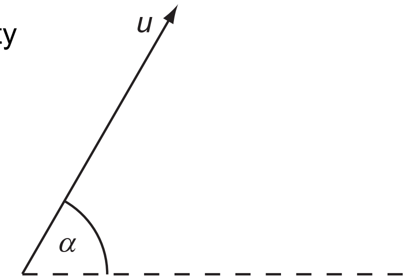


A 2.4 N **B** 5.3 N

C 6.7 N **D** 9.6 N

25 **9702/01/M/J/03/Q7**

A projectile is fired at an angle α to the horizontal at a speed u, as shown.



What are the vertical and horizontal components of its velocity after a time t ?

Assume that air resistance is negligible. The acceleration of free fall is g.

	vertical component	horizontal component
A	$u \sin \alpha$	$u \cos \alpha$
B	$u \sin \alpha - gt$	$u \cos \alpha - gt$
C	$u \sin \alpha - gt$	$u \cos \alpha$
D	$u \cos \alpha$	$u \sin \alpha - gt$

26 **9702/01/O/N/06/Q10***

A force F is applied to a freely moving object. At one instant of time, the object has velocity v and acceleration a.

Which quantities **must** be in the same direction?

A a and v only **B** a and F only **C** v and F only **D** v, F and a

27 **9702/01/O/N/06/Q11**

The diagram shows two identical spheres X and Y.

Initially X moves with speed v directly towards Y. Y is stationary. The spheres collide elastically.

What happens?

	X	Y
A	moves with speed $\frac{1}{2}v$ to the right	moves with speed $\frac{1}{2}v$ to the right
B	moves with speed v to the left	remains stationary
C	moves with speed $\frac{1}{2}v$ to the left	moves with speed $\frac{1}{2}v$ to the right
D	stops	moves with speed v to the right

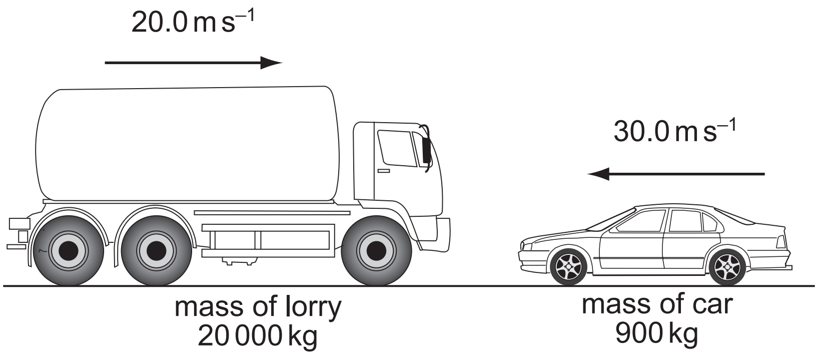
28 **9702/01/M/J/07/Q11**

A lorry of mass 20 000 kg is travelling at $20.0\,\mathrm{m\,s^{-1}}$. A car of mass 900 kg is travelling at $30.0\,\mathrm{m\,s^{-1}}$ towards the lorry.

What is the magnitude of the total momentum?

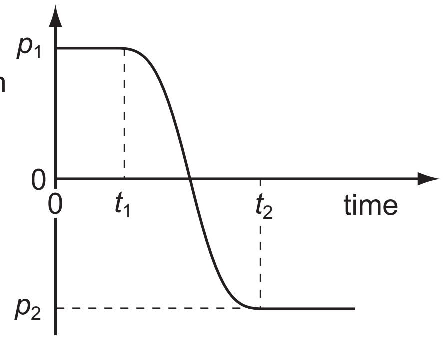
A 209 kN s **B** 373 kN s

C 427 kN s **D** 1045 kN s



29 **9702/01/M/J/07/Q10**

The graph shows the variation with time of the momentum of a ball as it is kicked in a straight line.



Initially, the momentum is p_1 at time t_1. At time t_2 the momentum is p_2.

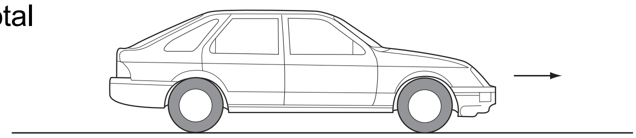
What is the magnitude of the average force acting on the ball between times t_1 and t_2?

A $\dfrac{p_1 - p_2}{t_2}$ **B** $\dfrac{p_1 - p_2}{t_2 - t_1}$

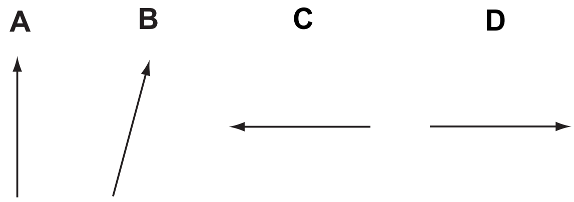
C $\dfrac{p_1 + p_2}{t_2}$ **D** $\dfrac{p_1 + p_2}{t_2 - t_1}$

30 **9702/01/O/N/07/Q11**

A car with front-wheel drive accelerates in the direction shown.

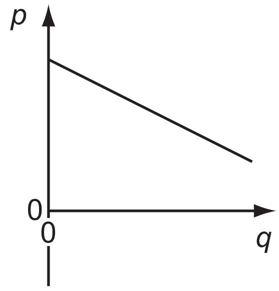


Which diagram best shows the direction of the total force exerted by the road on the front wheels?



31 **9702/01/O/N/07/Q12**

The graph shows how a certain quantity p varies with another quantity q for a parachutist falling at terminal speed.



What are the quantities p and q, and what is represented by the magnitude of the gradient of the graph?

	quantity p	quantity q	magnitude of gradient
A	height	time	terminal speed
B	momentum	time	weight of parachutist
C	height	potential energy	mass of parachutist
D	velocity	time	acceleration of free fall

32 **9702/01/M/J/08/Q9**

Which is a statement of the principle of conservation of momentum?

A Momentum is the product of mass and velocity.

B Momentum is conserved only in elastic collisions.

C Momentum is conserved by all bodies in a collision.

D Momentum is conserved providing no external forces act.

33 **9702/01/M/J/08/Q12**

A ball is falling at terminal speed in still air. The forces acting on the ball are upthrust, viscous drag and weight.

What is the order of increasing magnitude of these three forces?

A upthrust → viscous drag → weight

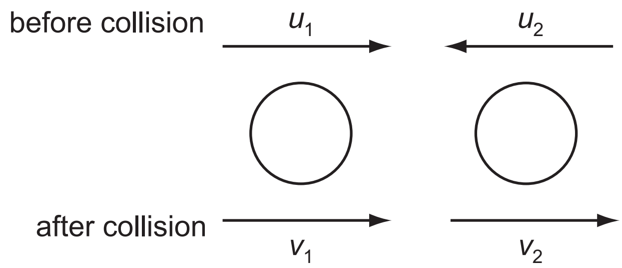
B viscous drag → upthrust → weight

C viscous drag → weight → upthrust

D weight → upthrust → viscous drag

34 9702/01/O/N/08/Q10

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before collision, and v_1 and v_2 after collision, in the directions shown below.



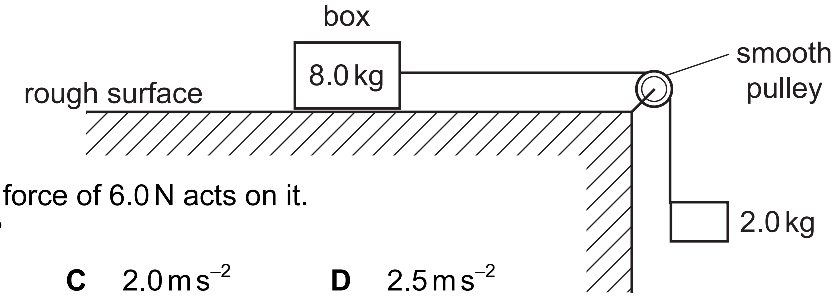
Which equation is correct if the collision is perfectly elastic?

A $u_1 - u_2 = v_2 + v_1$ **B** $u_1 - u_2 = v_2 - v_1$

C $u_1 + u_2 = v_2 + v_1$ **D** $u_1 + u_2 = v_2 - v_1$

35 9702/01/O/N/08/Q11

A box of mass 8.0 kg rests on a horizontal, rough surface. A string attached to the box passes over a smooth pulley and supports a 2.0 kg mass at its other end.



When the box is released, a friction force of 6.0 N acts on it.
What is the acceleration of the box?

A $1.4\,m\,s^{-2}$ **B** $1.7\,m\,s^{-2}$ **C** $2.0\,m\,s^{-2}$ **D** $2.5\,m\,s^{-2}$

36 9702/01/M/J/09/Q7

Which statement about Newton's laws of motion is correct?

A The first law follows from the second law.

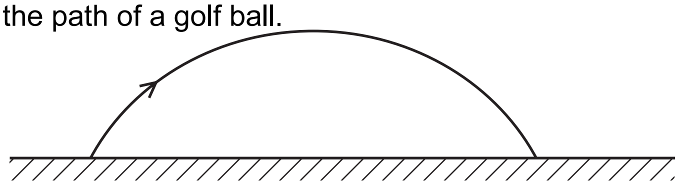
B The third law follows from the second law.

C Conservation of energy is a consequence of the third law.

D Conservation of linear momentum is a consequence of the first law.

37 9702/01/M/J/09/Q8

The diagram shows the path of a golf ball.

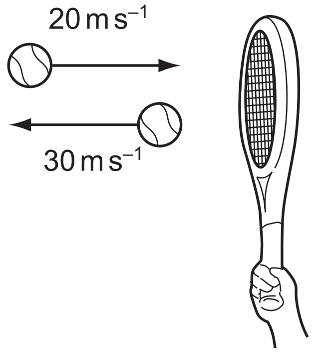


Which row describes changes in the horizontal and vertical components of the golf ball's velocity, when air resistance forces are ignored?

	horizontal	vertical
A	constant deceleration	constant acceleration downwards
B	constant deceleration	acceleration decreases upwards then increases downwards
C	constant velocity	constant acceleration downwards
D	constant velocity	acceleration decreases upwards then increases downwards

38 **9702/01/M/J/09/Q9**

A tennis ball of mass 100 g is struck by a tennis racket. The velocity of the ball is changed as shown.



What is the magnitude of the change in momentum of the ball?

A $1\,kg\,m\,s^{-1}$ **B** $5\,kg\,m\,s^{-1}$

C $1000\,kg\,m\,s^{-1}$ **D** $5000\,kg\,m\,s^{-1}$

39 **9702/01/M/J/09/Q12**

An object, made from two equal masses joined by a light rod, falls with uniform speed through air. The rod remains horizontal.

Which statement about the equilibrium of the system is correct?

A It is not in equilibrium because it is falling steadily.

B It is not in equilibrium because it is in motion.

C It is not in equilibrium because there is a resultant torque.

D It is in equilibrium because there is no resultant force and no resultant torque.

40 **9702/11/O/N/15/Q12**

Which statement about a ball that strikes a tennis racket and rebounds is **always** correct?

A Total kinetic energy of the ball is conserved.

B Total kinetic energy of the system is conserved.

C Total momentum of the ball is conserved.

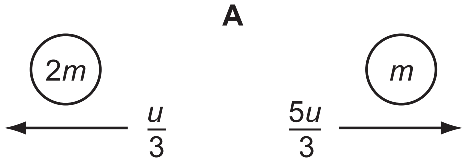
D Total momentum of the system is conserved.

41 **9702/13/O/N/12/Q11***

The diagram shows two spherical masses approaching each other head-on at an equal speed u. One has mass $2m$ and the other has mass m.

Which diagram, showing the situation after the collision, shows the result of an elastic collision?

A



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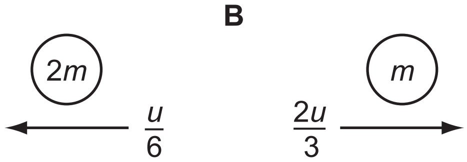
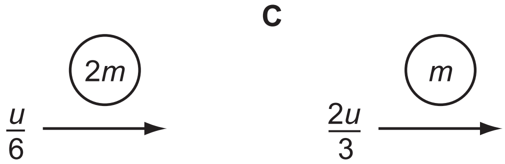
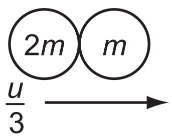


C



D



the spheres stick together

42 **9702/12/O/N/09/Q9**

A supermarket trolley, total mass 30 kg, is moving at $3.0\,m\,s^{-1}$. A retarding force of 60 N is applied to the trolley for 0.50 s in the opposite direction to the trolley's initial velocity.

What is the trolley's new velocity after the application of the force?

A $1.0\,m\,s^{-1}$ **B** $1.5\,m\,s^{-1}$ **C** $2.0\,m\,s^{-1}$ **D** $2.8\,m\,s^{-1}$

43 **9702/12/O/N/09/Q10**

What is the centre of gravity of an object?

A the geometrical centre of the object

B the point about which the total torque is zero

C the point at which the weight of the object may be considered to act

D the point through which gravity acts

44 **9702/12/M/J/10/Q10**

The diagram shows two identical spheres X and Y.

Initially, X moves with speed v directly towards Y. Y is stationary. The spheres collide elastically.

What happens?

	X	Y
A	moves with speed $\frac{1}{2}v$ to the right	moves with speed $\frac{1}{2}v$ to the right
B	moves with speed v to the left	remains stationary
C	moves with speed $\frac{1}{2}v$ to the left	moves with speed $\frac{1}{2}v$ to the right
D	stops	moves with speed v to the right

45 **9702/12/M/J/10/Q11**

Two equal masses travel towards each other on a frictionless air track at speeds of $60\,cm\,s^{-1}$ and $40\,cm\,s^{-1}$. They stick together on impact.

What is the speed of the masses after impact?

A $10\,cm\,s^{-1}$ **B** $20\,cm\,s^{-1}$

C $40\,cm\,s^{-1}$ **D** $50\,cm\,s^{-1}$

46 **9702/11/O/N/10/Q10**

The gravitational field strength on the surface of planet P is one tenth of that on the surface of planet Q.

On the surface of P, a body has a mass of 1.0 kg and a weight of 1.0 N.

What are the mass and weight of the same body on the surface of planet Q?

	mass on Q / kg	weight on Q / N
A	1.0	0.1
B	1.0	10
C	10	10
D	10	100

47 9702/11/O/N/10/Q11

A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\frac{v_1}{v_2}$?

A $\frac{M_1}{M_2}$ **B** $\frac{M_2}{M_1}$ **C** $\sqrt{\frac{M_1}{M_2}}$ **D** $\sqrt{\frac{M_2}{M_1}}$

48 9702/11/O/N/10/Q12

Two experiments are carried out using two trolleys of equal mass. All moving parts of the trolleys are frictionless, as is the surface that the trolleys move over. In both experiments, trolley X moves towards trolley Y, which is initially stationary.

After the collision in experiment 1, X is stationary and Y moves off to the right.

After the collision in experiment 2, the trolleys join and move off together.

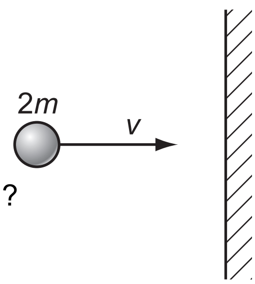
What types of collision occur in these experiments?

	experiment 1	experiment 2
A	elastic	elastic
B	elastic	inelastic
C	inelastic	elastic
D	inelastic	inelastic

49 9702/12/O/N/10/Q9

A particle of mass $2m$ and velocity v strikes a wall.

The particle rebounds along the same path after colliding with the wall. The collision is inelastic.



What is a possible change in the momentum of the ball during the collision?

A mv **B** $2mv$ **C** $3mv$ **D** $4mv$

50 9702/12/O/N/10/Q10

Which defines the weight of a body?

A the amount of matter in the body

B the force of gravity on the body

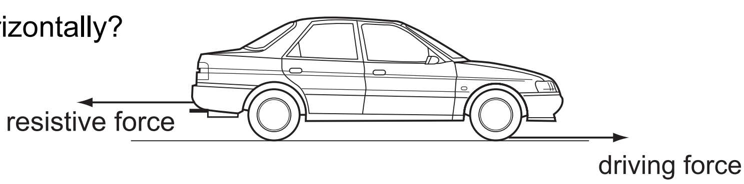
C the number of particles in the body

D the product of the body's volume and density

51 9702/01/M/J/08/Q11

A car of mass 750 kg has a horizontal driving force of 2.0 kN acting on it. It has a forward horizontal acceleration of $2.0\,m\,s^{-2}$.

What is the resistive force acting horizontally?

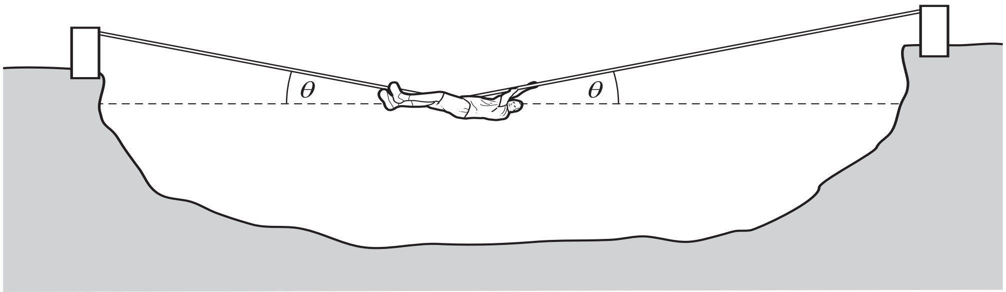


A 0.5 kN **B** 1.5 kN

C 2.0 kN **D** 3.5 kN

52 **9702/12/O/N/10/Q11**

The diagram shows a rope bridge that a student makes on an adventure training course. The student has a weight W.



Which formula gives the tension T in the rope?

A $T = \dfrac{W}{2\cos\theta}$ **B** $T = \dfrac{W}{2\sin\theta}$ **C** $T = \dfrac{W}{\cos\theta}$ **D** $T = \dfrac{W}{\sin\theta}$

53 **9702/11/M/J/11/Q8**

A body has a weight of 58.9 N when on the Earth. On the Moon, the acceleration of free fall is $1.64\,\text{m}\,\text{s}^{-2}$.

What are the weight and the mass of the body when it is on the Moon?

	weight / N	mass / kg
A	9.85	1.00
B	9.85	6.00
C	58.9	1.00
D	58.9	6.00

54 **9702/11/M/J/11/Q9**

A body of mass m, moving at velocity v, collides with a stationary body of the same mass and sticks to it.

Which row describes the momentum and kinetic energy of the two bodies after the collision?

	momentum	kinetic energy
A	mv	$\frac{1}{4}mv^2$
B	mv	$\frac{1}{8}mv^2$
C	$2mv$	$\frac{1}{2}mv^2$
D	$2mv$	mv^2

55 **9702/11/M/J/11/Q10**

A molecule of mass m travelling horizontally with velocity u hits a vertical wall at right-angles to its velocity. It then rebounds horizontally with the same speed.

What is its change in momentum?

A zero **B** mu **C** $-mu$ **D** $-2mu$

56 **9702/12/M/J/11/Q11**

The momentum of an object changes from $160\,\text{kg}\,\text{m}\,\text{s}^{-1}$ to $240\,\text{kg}\,\text{m}\,\text{s}^{-1}$ in 2 s.

What is the mean resultant force on the object during the change?

A 40 N **B** 80 N **C** 200 N **D** 400 N

57 **9702/12/M/J/11/Q10**

A force F is applied to a freely moving object. At one instant of time, the object has velocity v and acceleration a.

Which quantities **must** be in the same direction?

A a and v only **B** a and F only **C** v and F only **D** v, F and a

58 **9702/12/M/J/11/Q12**

A car accelerates in a straight line.

A graph of the momentum of the car is plotted against time.

What is evaluated by finding the gradient of the graph at a particular time?

A the acceleration of the car

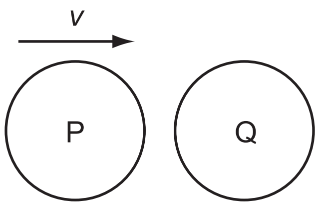
B the resultant force on the car

C the kinetic energy of the car

D the power supplied to the car

59 **9702/12/M/J/11/Q13**

The diagram shows a particle P, travelling at speed v, about to collide with a stationary particle Q of the same mass. The collision is perfectly elastic.



Which statement describes the motion of P and of Q immediately after the collision?

A P rebounds with speed $\frac{1}{2}v$ and Q acquires speed $\frac{1}{2}v$.

B P rebounds with speed v and Q remains stationary.

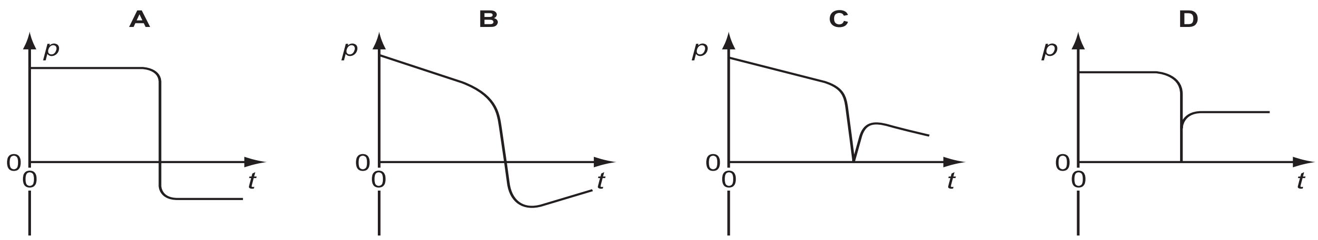
C P and Q both travel in the same direction with speed $\frac{1}{2}v$.

D P comes to a standstill and Q acquires speed v.

60 **9702/11/O/N/11/Q12**

An ice-hockey puck slides along a horizontal, frictionless ice-rink surface. It collides inelastically with a wall at right angles to its path, and then rebounds along its original path.

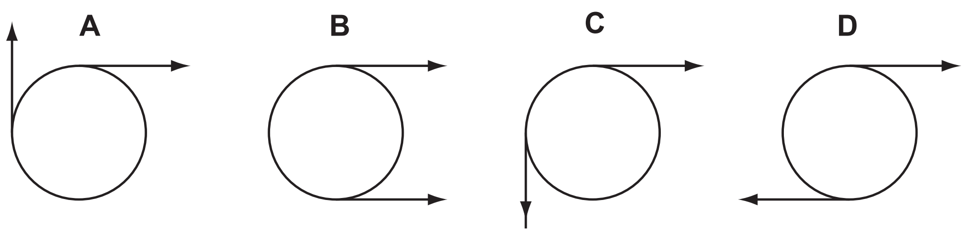
Which graph shows the variation with time t of the momentum p of the puck?



61 **9702/12/O/N/11/Q13**

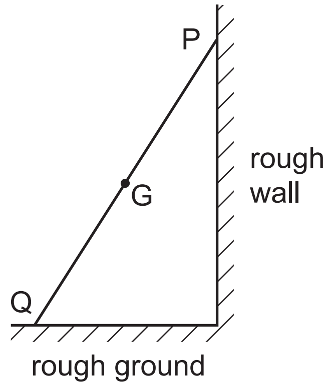
Two co-planar forces act on the rim of a wheel. The forces are equal in magnitude.

Which arrangement of forces provides only a couple?



62 **9702/11/O/N/11/Q13**

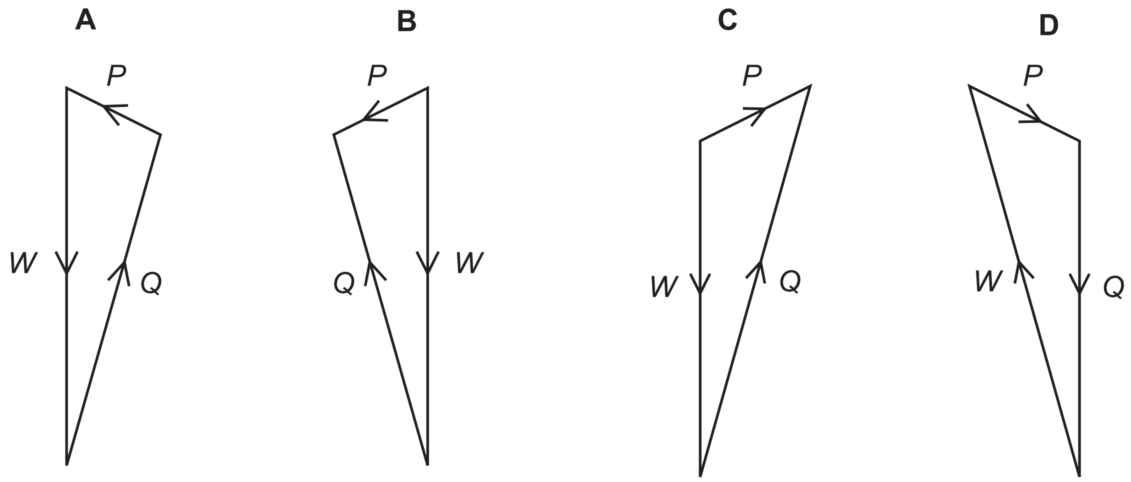
A ladder rests in equilibrium on rough ground against a rough wall.



Its weight W acts through the centre of gravity G. Forces also act on the ladder at P and at Q.

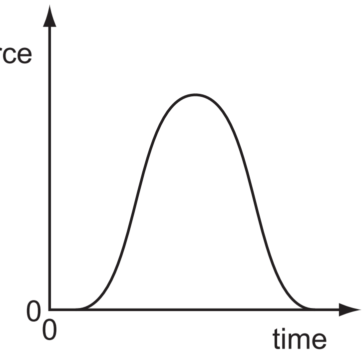
These forces are *P* and *Q* respectively.

Which vector triangle represents the forces on the ladder?



63 **9702/11/O/N/15/Q10**

A golf ball is hit by a club. The graph shows the variation with time of the force exerted on the ball by the club.



Which quantity, for the time of contact, **cannot** be found from the graph?

A the average force on the ball

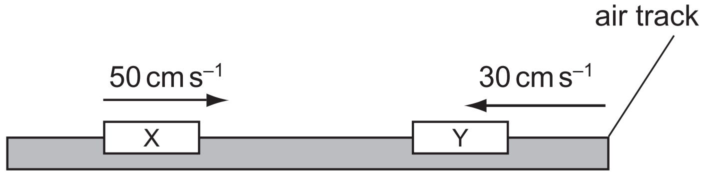
B the change in momentum of the ball

C the contact time between the ball and the club

D the maximum acceleration of the ball

64 **9702/13/O/N/15/Q12**

Two equal masses X and Y are moving towards each other on a frictionless air track as shown. The masses make an elastic collision.



Which row gives possible velocities for the two masses after the collision?

	velocity of X	velocity of Y
A	zero	$20\,\text{cm}\,\text{s}^{-1}$ to the right
B	$10\,\text{cm}\,\text{s}^{-1}$ to the right	$10\,\text{cm}\,\text{s}^{-1}$ to the right
C	$20\,\text{cm}\,\text{s}^{-1}$ to the left	zero
D	$30\,\text{cm}\,\text{s}^{-1}$ to the left	$50\,\text{cm}\,\text{s}^{-1}$ to the right

65 **9702/12/O/N/11/Q10**

A group of students investigating the principle of conservation of momentum use a small truck travelling over a frictionless surface.

Sand is dropped into the truck as it passes X. At Y, a trapdoor in the bottom of the truck opens and the sand falls out.

How does the velocity of the truck change when the sand is added to the truck at X and then leaves the truck at Y?

	at X	at Y
A	decreases	increases
B	decreases	stays the same
C	stays the same	increases
D	stays the same	stays the same

66 **9702/12/O/N/11/Q11**

An object of mass 20 kg is travelling at a constant speed of $6.0\,\mathrm{m\,s^{-1}}$.

It collides with an object of mass 12 kg travelling at a constant speed of $15\,\mathrm{m\,s^{-1}}$ in the opposite direction. The objects stick together.

What is the speed of the objects immediately after the collision?

A $1.9\,\mathrm{m\,s^{-1}}$ **B** $9.0\,\mathrm{m\,s^{-1}}$ **C** $9.4\,\mathrm{m\,s^{-1}}$ **D** $21\,\mathrm{m\,s^{-1}}$

67 **9702/01/O/N/08/Q9**

A ball falls vertically and bounces on the ground.

The following statements are about the forces acting while the ball is in contact with the ground.

Which statement is correct?

A The force that the ball exerts on the ground is always equal to the weight of the ball.

B The force that the ball exerts on the ground is always equal in magnitude and opposite in direction to the force the ground exerts on the ball.

C The force that the ball exerts on the ground is always less than the weight of the ball.

D The weight of the ball is always equal in magnitude and opposite in direction to the force that the ground exerts on the ball.

68 **9702/11/M/J/16/Q10**

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before collision. After the collision, the spheres separate with speeds v_1 and v_2 in the directions shown below.

Which equation must be correct if the collision is perfectly elastic?

A $u_1 - u_2 = v_2 + v_1$ **C** $u_1 + u_2 = v_2 + v_1$

B $u_1 - u_2 = v_2 - v_1$ **D** $u_1 + u_2 = v_2 - v_1$

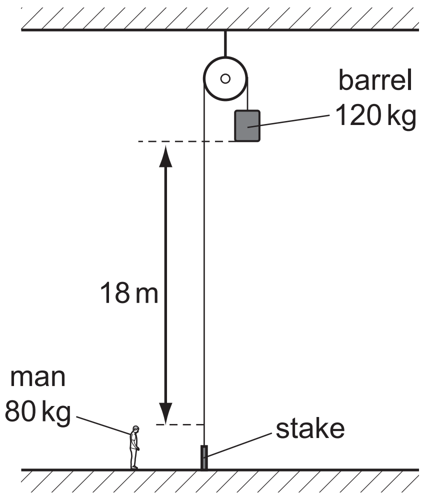
69 9702/12/M/J/12/Q11

Which row correctly states whether momentum and kinetic energy are conserved in an inelastic collision in which there are no external forces?

	momentum	kinetic energy
A	conserved	conserved
B	conserved	not conserved
C	not conserved	conserved
D	not conserved	not conserved

70 9702/13/M/J/12/Q10

The diagram shows a barrel suspended from a frictionless pulley on a building. The rope supporting the barrel goes over the pulley and is secured to a stake at the bottom of the building.



A man stands close to the stake. The bottom of the barrel is 18 m above the man's head. The mass of the barrel is 120 kg and the mass of the man is 80 kg.

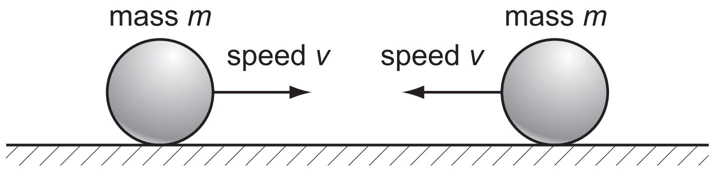
The man keeps hold of the rope after untying it from the stake and is lifted upwards as the barrel falls.

What is the man's upward speed when his head is level with the bottom of the barrel? (Use $g = 10\,\text{m}\,\text{s}^{-2}$.)

A $6\,\text{m}\,\text{s}^{-1}$ **B** $8\,\text{m}\,\text{s}^{-1}$ **C** $13\,\text{m}\,\text{s}^{-1}$ **D** $19\,\text{m}\,\text{s}^{-1}$

71 9702/12/O/N/12/Q13

Two identical, perfectly elastic spheres have the same mass *m*. They travel towards each other with the same speed *v* along a horizontal frictionless surface.



Which statement about the sum of the kinetic energies of the spheres is correct?

A The sum of their kinetic energies before impact is zero.

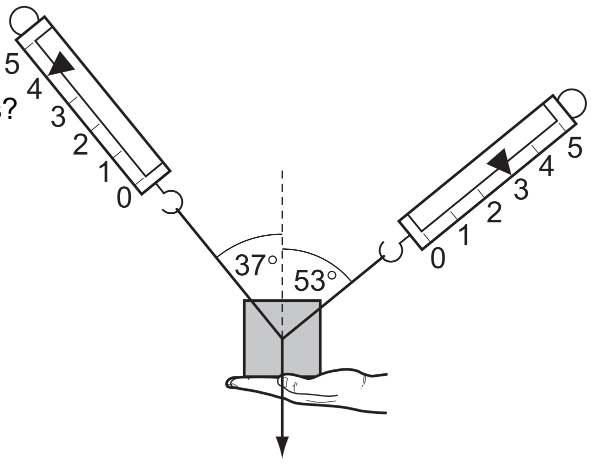
B The sum of their kinetic energies before impact is $\frac{1}{2}mv^2$.

C The sum of their kinetic energies after impact is zero.

D The sum of their kinetic energies after impact is mv^2.

72 9702/12/M/J/12/Q14

A 1.2 kg mass is supported by a person's hand and two newton-meters as shown.



When the person's hand is removed,
What is the initial vertical acceleration of the mass?

A $0.6\,\mathrm{m\,s^{-2}}$

B $2\,\mathrm{m\,s^{-2}}$

C $4\,\mathrm{m\,s^{-2}}$

D $6\,\mathrm{m\,s^{-2}}$

73 9702/12/O/N/12/Q15

A lorry of mass 20 000 kg has a constant resultant force F acting on it.
It accelerates from $6.0\,\mathrm{m\,s^{-1}}$ to $30.0\,\mathrm{m\,s^{-1}}$ in a time of 300 s.
What is the change in momentum of the lorry and the value of F?

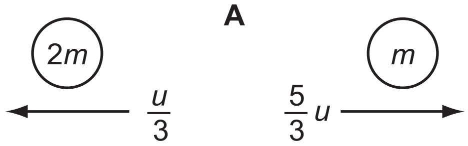
	change in momentum / N s	force F / N
A	48 000	160
B	480 000	1600
C	600 000	2000
D	600 000	20 000

74 9702/13/O/N/12/Q11

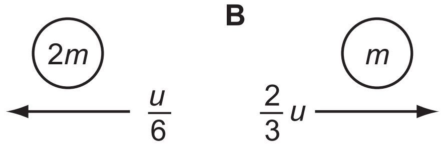
The diagram shows two spherical masses approaching each other head-on at an equal speed u. One is of mass m and the other of mass $2m$.

Which diagram, showing the situation after the collision, is **not** consistent with the principle of conservation of momentum?

A

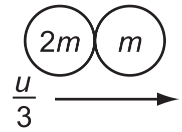


B



C

D



the spheres stick together

75 9702/13/O/N/12/Q11*

A molecule of mass m travelling at speed v hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the kinetic energy and in the momentum of the molecule caused by the collision?

	change in momentum	change in kinetic energy
A	0	0
B	0	mv^2
C	$2mv$	0
D	mv^2	0

76 9702/13/O/N/12/Q13

The IKAROS satellite has mass 320 kg and moves through space using a solar sail of area 20 m^2. The average solar wind pressure is 1.0×10^{-5} N m^{-2}.

What is the acceleration of the satellite caused by the solar wind?

A 3.1×10^{-8} m s^{-2}

B 6.3×10^{-7} m s^{-2}

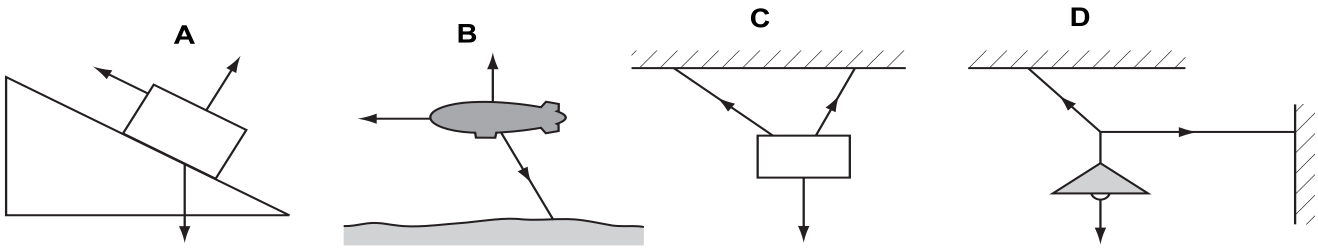
C 3.2×10^{-3} m s^{-2}

D 6.4×10^{-2} m s^{-2}

77 9702/13/O/N/12/Q17

The diagrams show the forces acting on different bodies.

Which body **cannot** be in equilibrium?



78 9702/1/M/J/02/Q9

Two similar spheres, each of mass m and travelling with speed v, are moving towards each other.

The spheres have a head-on elastic collision.

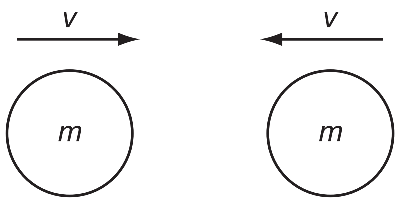
Which statement is correct?

A The spheres stick together on impact.

B The total kinetic energy after impact is mv^2.

C The total kinetic energy before impact is zero.

D The total momentum before impact is $2mv$.



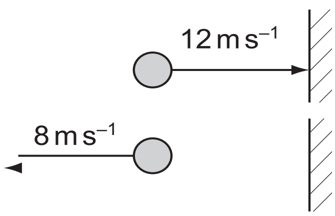
79 9702/12/O/N/12/Q12

A ball of mass 0.5 kg is thrown against a wall at a speed of 12 m s^{-1}. It bounces back with a speed of 8 m s^{-1}. The collision lasts for 0.10 s.

What is the average force on the ball due to the collision?

A 0.2 N **B** 1 N

C 20 N **D** 100 N



80 **9702/11/M/J/13/Q10**

Which of the following is a statement of the principle of conservation of momentum?

A In an elastic collision momentum is constant.

B Momentum is the product of mass and velocity.

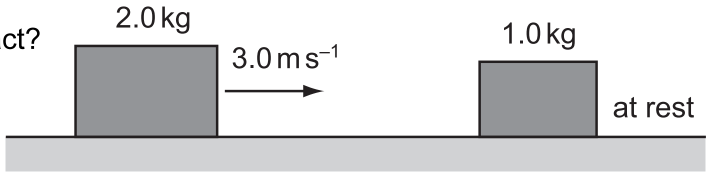
C The force acting on a body is proportional to its rate of change of momentum.

D The momentum of an isolated system is constant.

81 **9702/11/M/J/13/Q11**

A 2.0 kg mass travelling at $3.0\,\text{m}\,\text{s}^{-1}$ on a frictionless surface collides head-on with a stationary 1.0 kg mass. The masses stick together on impact.

How much kinetic energy is lost on impact?



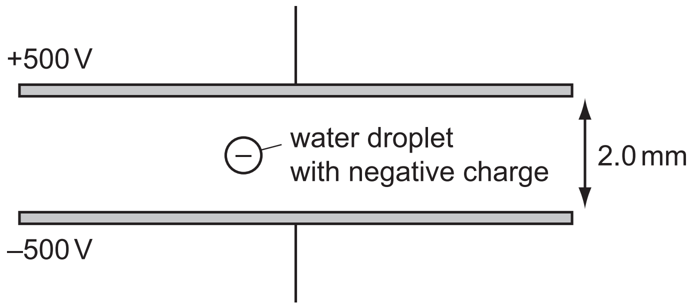
A zero **B** 2.0 J

C 2.4 J **D** 3.0 J

82 **9702/11/M/J/13/Q13**

A small water droplet of mass $3.0\,\mu\text{g}$ carries a charge of $-6.0 \times 10^{-11}\,\text{C}$. The droplet is situated in the Earth's gravitational field between two horizontal metal plates. The potential of the upper plate is +500 V and the potential of the lower plate is –500 V.

What is the motion of the droplet?



A It accelerates downwards.

B It remains stationary.

C It accelerates upwards.

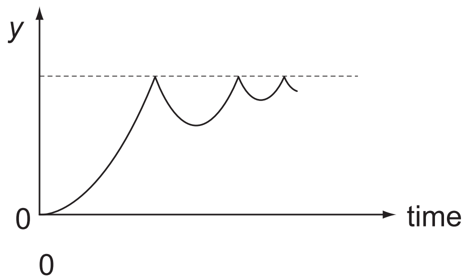
D It moves upwards at a constant velocity.

83 **9702/12/M/J/13/Q8**

A ball is released from rest above a horizontal surface and bounces several times.

The graph shows how, for this ball, a quantity y varies with time.

What is the quantity y?



A acceleration

B displacement

C kinetic energy

D velocity

84 **9702/12/M/J/13/Q9**

A strong wind of speed $33\,\text{m}\,\text{s}^{-1}$ blows against a wall. The density of the air is $1.2\,\text{kg}\,\text{m}^{-3}$. The wall has an area of $12\,\text{m}^2$ at right angles to the wind velocity. The air has its speed reduced to zero when it hits the wall.

What is the approximate force exerted by the air on the wall?

A 330 N **B** 400 N **C** 480 N **D** 16 000 N

85 **9702/12/O/N/13/Q10**

An astronaut of mass m in a spacecraft experiences a gravitational force $F = mg$ when stationary on the launchpad.

What is the gravitational force on the astronaut when the spacecraft is launched vertically upwards with an acceleration of $0.2g$?

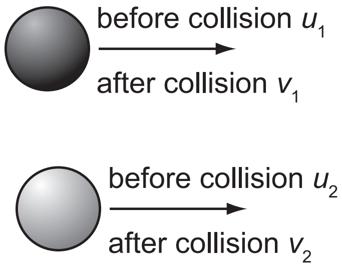
A $1.2mg$ **B** mg **C** $0.8mg$ **D** 0

86 9702/11/O/N/16/Q13

Two spheres travel along the same line with velocities u_1 and u_2. They collide and after collision their velocities are v_1 and v_2.

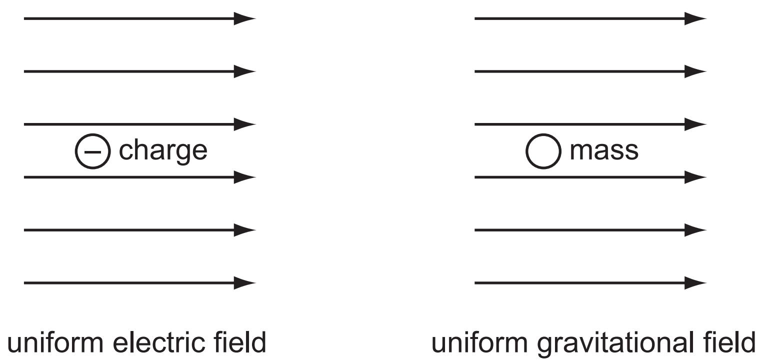
Which collision is **not** elastic?

	u_1 / m s^{-1}	u_2 / m s^{-1}	v_1 / m s^{-1}	v_2 / m s^{-1}
A	2	–5	–5	–2
B	3	–3	0	6
C	3	–2	1	6
D	5	2	3	6

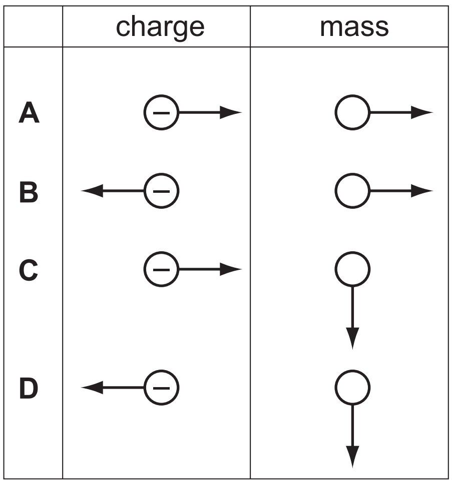


87 9702/13/M/J/13/Q12

The diagrams show a negative electric charge situated in a uniform electric field and a mass situated in a uniform gravitational field.



Which row shows the directions of the forces acting on the charge and on the mass?



88 9702/12/M/J/13/Q10

Two bodies travelling in a straight line collide in a perfectly elastic collision. Which of the following statements **must** be correct?

A The initial speed of one body will be the same as the final speed of the other body.

B The relative speed of approach between the two bodies equals their relative speed of separation.

C The total momentum is conserved but the total kinetic energy will be reduced.

D One of the bodies will be stationary at one instant.

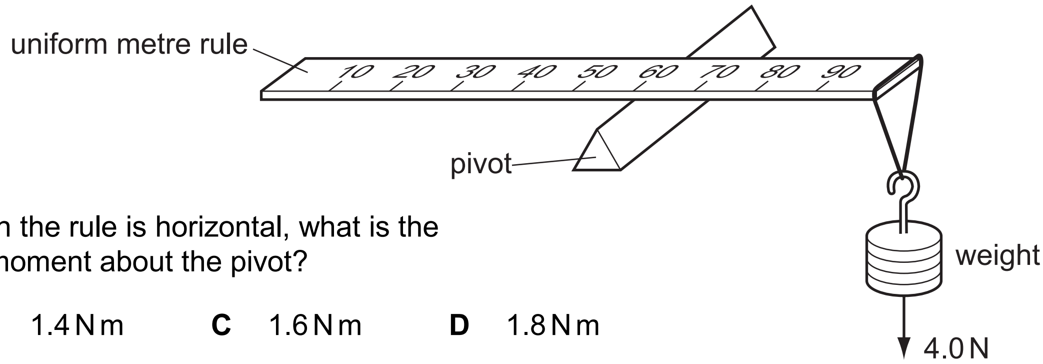
89 **9702/12/O/N/13/Q9**

What is meant by the mass and by the weight of an object on the Earth?

	mass	weight
A	its momentum divided by its velocity	the work done in lifting it one metre
B	the gravitational force on it	the property that resists its acceleration
C	the pull of the Earth on it	its mass divided by the acceleration of free fall
D	the property that resists its acceleration	the pull of the Earth on it

90 **9702/12/O/N/13/Q15**

A uniform metre rule of weight 2.0 N is pivoted at the 60 cm mark. A 4.0 N weight is suspended from one end, causing the rule to rotate about the pivot.



At the instant when the rule is horizontal, what is the resultant turning moment about the pivot?

A zero **B** 1.4 N m **C** 1.6 N m **D** 1.8 N m

91 **9702/11/M/J/14/Q9**

An object of mass 4.0 kg moving with a speed of $3.0\,\mathrm{m\,s^{-1}}$ strikes a stationary object in an inelastic collision.

Which statement is correct?

A After collision, the total kinetic energy is 18 J.

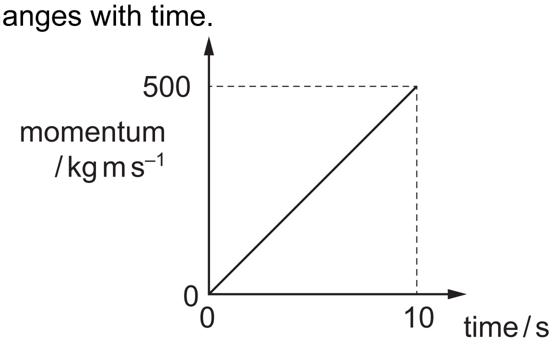
B After collision, the total kinetic energy is less than 18 J.

C Before collision, the total kinetic energy is 12 J.

D Before collision, the total kinetic energy is less than 12 J.

92 **9702/11/M/J/14/Q10**

The graph shows how the momentum of a motorcycle changes with time.



What is the resultant force on the motorcycle?

A 50 N **B** 500 N

C 2500 N **D** 5000 N

93 **9702/11/O/N/14/Q9**

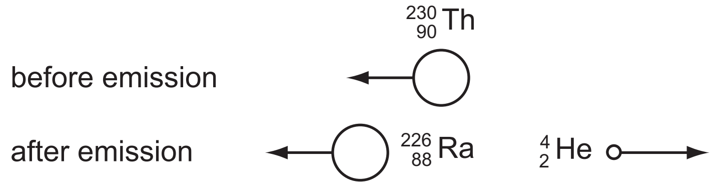
Two railway trucks of masses m and $3m$ move towards each other in opposite directions with speeds $2v$ and v respectively. These trucks collide and stick together.

What is the speed of the trucks after the collision?

A $\frac{v}{4}$ **B** $\frac{v}{2}$ **C** v **D** $\frac{5v}{4}$

94 **9702/13/O/N/13/Q10**

A moving thorium nucleus ${}^{230}_{90}Th$ spontaneously emits an α-particle. The nucleus formed is a radium nucleus ${}^{226}_{88}Ra$, as shown.



Which statement is correct?

A The kinetic energy of the α-particle equals the kinetic energy of the radium nucleus.

B The momentum of the α-particle equals the momentum of the radium nucleus.

C The total momentum before the emission equals the total momentum after the emission.

D The velocity of the α-particle equals the velocity of the radium nucleus.

95 **9702/13/O/N/13/Q11**

An isolated system consists of two bodies on which no external forces act. The two bodies collide with each other and stick together on impact.

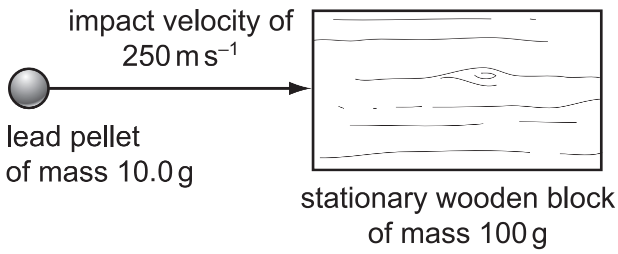
Which row correctly compares the total kinetic energy and the total momentum of the bodies before and after the collision?

	total kinetic energy before and after the collision	total momentum before and after the collision
A	different	different
B	different	the same
C	the same	different
D	the same	the same

96 **9702/13/O/N/13/Q13**

A lead pellet of mass 10.0 g is shot horizontally into a stationary wooden block of mass 100 g. The pellet hits the block with an impact velocity of $250\,m\,s^{-1}$. It embeds itself in the block and it does not emerge.

What will be the speed of the block immediately after the pellet is embedded?

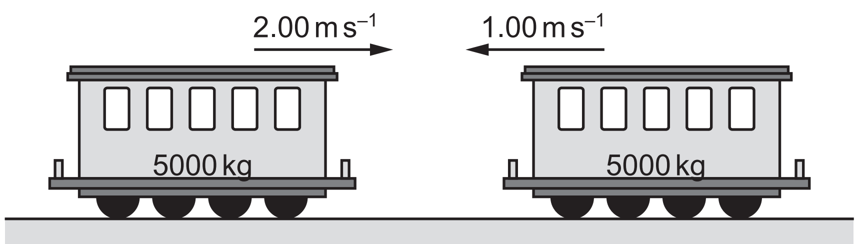


A $23\,m\,s^{-1}$ **B** $25\,m\,s^{-1}$

C $75\,m\,s^{-1}$ **D** $79\,m\,s^{-1}$

97 **9702/12/M/J/14/Q7**

Two train carriages each of mass 5000 kg roll toward one another on a level track. One is travelling at $2.00\,m\,s^{-1}$ and the other at $1.00\,m\,s^{-1}$, as shown.



They collide and join together.

What is the kinetic energy lost during the collision?

A 1250 J **B** 7500 J

C 11 250 J **D** 12 500 J

98 **9702/12/M/J/14/Q8**

A resultant force causes a body to accelerate.

What is equal to the resultant force?

A the acceleration of the body per unit mass

B the change in kinetic energy of the body per unit time

C the change in momentum of the body per unit time

D the change in velocity of the body per unit time

99 **9702/12/M/J/14/Q9**

A ship of mass 8.4×10^7 kg is approaching a harbour with speed $16.4\,\text{m}\,\text{s}^{-1}$. By using reverse thrust it can maintain a constant total stopping force of 920 000 N.

How long will it take to stop?

A 15 seconds **B** 150 seconds **C** 25 minutes **D** 250 minutes

100 **9702/11/O/N/14/Q7**

What is the principle of conservation of momentum?

A Force is equal to the rate of change of momentum.

B Momentum is the product of mass and velocity.

C The total momentum of a system remains constant provided no external force acts on it.

D The total momentum of two bodies after collision is equal to their total momentum before collision.

101 **9702/11/O/N/14/Q8**

Water is pumped through a hose-pipe at a rate of 90 kg per minute. It emerges from the hose-pipe horizontally with a speed of $20\,\text{m}\,\text{s}^{-1}$.

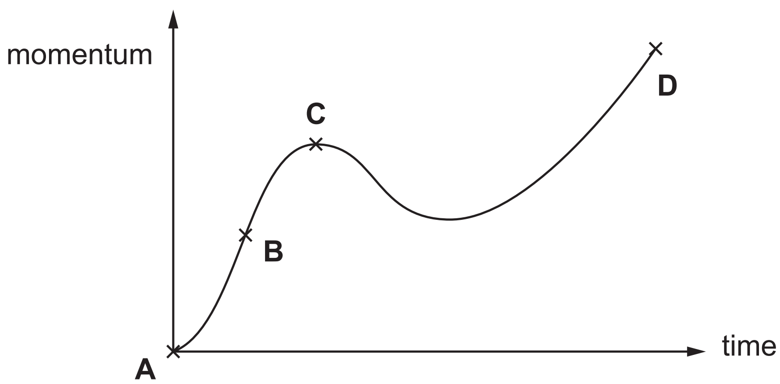
Which force is required from a person holding the hose-pipe to prevent it moving backwards?

A 30 N **B** 270 N **C** 1800 N **D** 10 800 N

102 **9702/13/O/N/14/Q10**

A body experiences a varying resultant force that causes its momentum to vary, as shown in the graph.

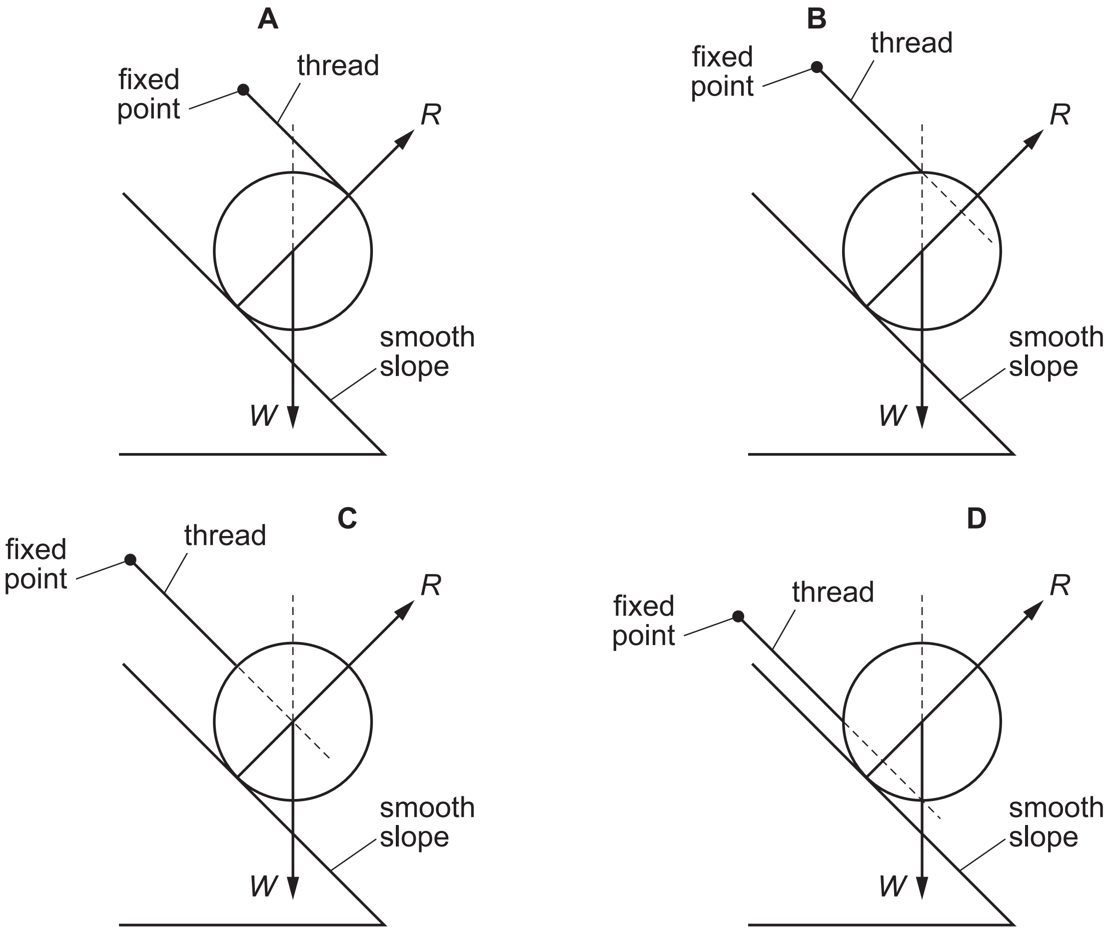
At which point does the resultant force have the largest value?



103 9702/13/O/N/14/Q13

A cylinder of weight W is placed on a smooth slope. The contact force of the slope on the cylinder is R. A thread is attached to the surface of the cylinder. The other end of the thread is fixed.

Which diagram shows the cylinder in equilibrium?



104 9702/12/M/J/15/Q11

Trolley X, moving along a horizontal frictionless track, collides with a stationary trolley Y. The two trolleys become attached and move off together.

Which statement about this interaction is correct?

A Some of the kinetic energy of trolley X is changed to momentum in the collision.

B Some of the momentum of trolley X is changed to kinetic energy in the collision.

C Trolley X loses some of its momentum as heat in the collision.

D Trolley X shares its momentum with trolley Y but some of its kinetic energy is lost.

105 9702/12/M/J/15/Q12

An astronaut throws a stone with a horizontal velocity near to the Moon's surface.

Which row describes the horizontal and vertical forces acting on the stone after release?

	horizontal force	vertical force
A	constant	constant
B	constant	decreasing
C	zero	constant
D	zero	decreasing

106 **9702/11/M/J/15/Q10**

What is a reasonable estimate of the average gravitational force acting on a fully grown woman standing on the Earth?

A 60 N **B** 250 N **C** 350 N **D** 650 N

107 **9702/13/O/N/12/Q12**

A molecule of mass m travelling at speed v hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the momentum and in the kinetic energy of the molecule caused by the collision?

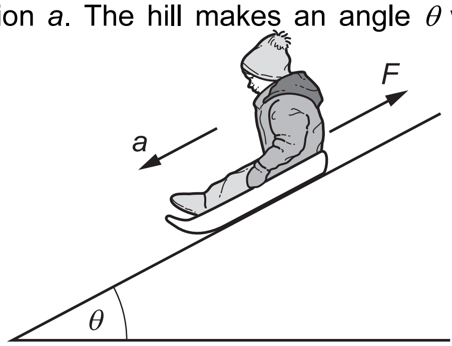
	change in momentum	change in kinetic energy
A	0	0
B	0	mv^2
C	$2mv$	0
D	mv^2	0

108 **9702/11/M/J/15/Q12**

A child on a sledge slides down a hill with acceleration a. The hill makes an angle θ with the horizontal.

The total mass of the child and the sledge is m

. The acceleration of free fall is g.



What is the friction force F?

A $m(g\cos\theta - a)$

B $m(g\cos\theta + a)$

C $m(g\sin\theta - a)$

D $m(g\sin\theta + a)$

109 **9702/12/M/J/17/Q8**

A golf ball of mass m is dropped onto a hard surface from a height h_1 and rebounds to a height h_2.

The momentum of the golf ball just as it reaches the surface is different from its momentum just as it leaves the surface.

What is the total change in the momentum of the golf ball between these two instants? (Ignore air resistance.)

A $m\sqrt{2gh_1} - m\sqrt{2gh_2}$

B $m\sqrt{2gh^1} + m\sqrt{2gh^2}$

C $m\sqrt{2g(h_1 - h_2)}$

D $m\sqrt{2g(h^1 + h^2)}$

110 **9702/12/M/J/15/Q13**

Newton's third law of motion is often summarised as 'Every action (force) has an equal and opposite reaction.'

A book rests on a table.

If the weight of the book is the 'action' force, what is the 'reaction' force?

A the pull of the book on the Earth

B the pull of the Earth on the book

C the push of the book on the table

D the push of the table on the book

111 **9702/13/M/J/15/Q11**

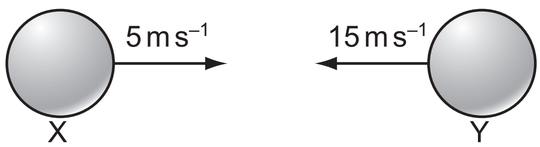
A moving object strikes a stationary object. The collision is inelastic. The objects move off together.

Which row shows the possible values of total momentum and total kinetic energy for the system before and after the collision?

	total momentum before collision / $kg\,m\,s^{-1}$	total momentum after collision / $kg\,m\,s^{-1}$	total kinetic energy before collision / J	total kinetic energy after collision / J
A	6	2	90	30
B	6	6	30	90
C	6	6	90	30
D	6	6	90	90

112 **9702/13/M/J/15/Q12**

Two balls X and Y are moving towards each other with speeds of $5\,m\,s^{-1}$ and $15\,m\,s^{-1}$ respectively.



They make a perfectly elastic head-on collision and ball Y moves to the right with a speed of $7\,m\,s^{-1}$.

What is the speed and direction of ball X after the collision?

A $3\,m\,s^{-1}$ to the left

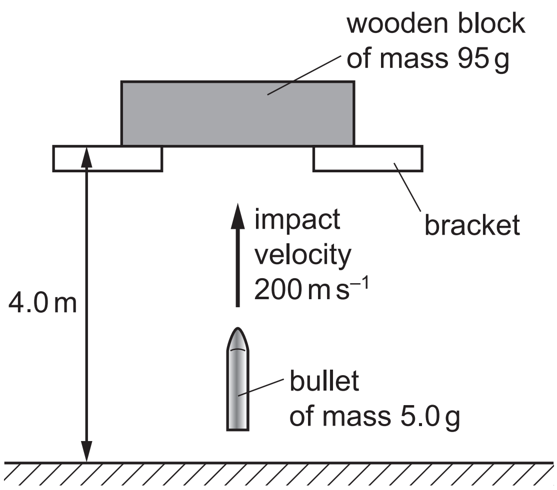
B $13\,m\,s^{-1}$ to the left

C $3\,m\,s^{-1}$ to the right

D $13\,m\,s^{-1}$ to the right

113 **9702/13/M/J/15/Q13**

A wooden block is freely supported on brackets at a height of 4.0 m above the ground, as shown.



A bullet of mass 5.0 g is shot vertically upwards into the wooden block of mass 95 g. It embeds itself in the block. The impact causes the block to rise above its supporting brackets.

The bullet hits the block with a velocity of $200\,m\,s^{-1}$. How far above the ground will the block be at the maximum height of its path?

A 5.1 m **B** 5.6 m **C** 9.1 m **D** 9.6 m

114 9702/11/M/J/17/Q8

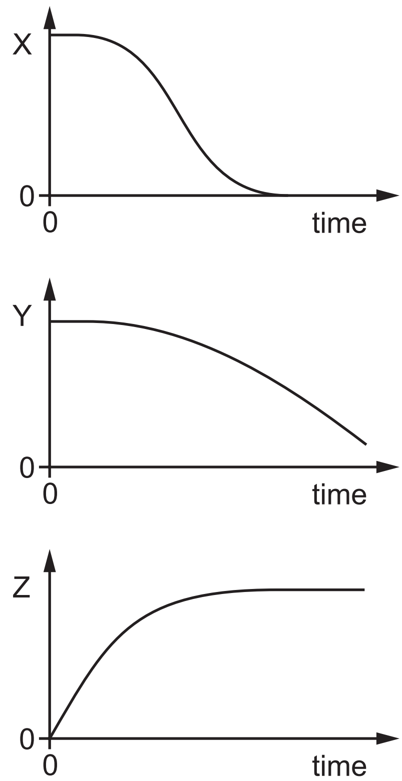
An object is dropped at time $t = 0$ from a high building. Air resistance is significant.

Three graphs are plotted against time.

the height of the object above the ground

the speed of the object

the magnitude of the resultant force on the object



What are the quantities X, Y and Z?

	height of the object above the ground	speed of the object	magnitude of the resultant force on the object
A	X	Y	Z
B	X	Z	Y
C	Y	Z	X
D	Z	Y	X

115 9702/11/M/J/17/Q7

The mass of a rocket-propelled truck is approximately equal to the mass of the fuel in its tank. The fuel is ignited and the truck is propelled along horizontal tracks by a constant force. The effect of air resistance is negligible.

During a test run the fuel is consumed at a constant rate.

Which statement describes the acceleration of the truck during the test run?

A The acceleration of the truck decreases as the fuel is consumed.

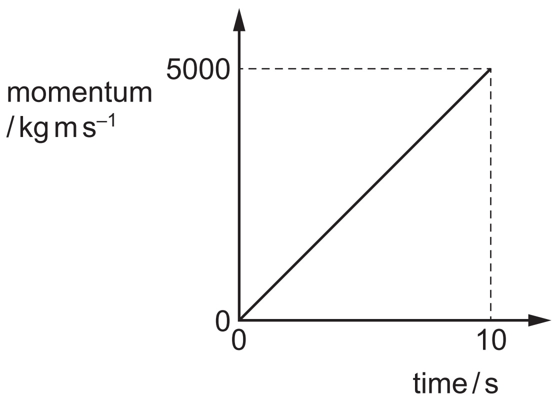
B The acceleration of the truck increases as the fuel is consumed.

C The acceleration of the truck remains constant.

D The acceleration of the truck is zero and the truck moves at a constant velocity.

116 9702/11/M/J/17/Q10

The graph shows how the momentum of a motorcycle changes with time.



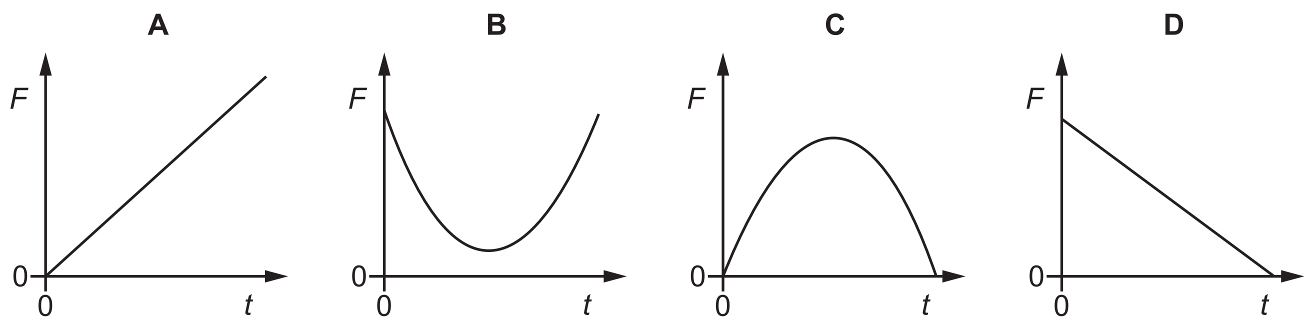
What is the resultant force on the motorcycle?

A 500 N **B** 5000 N **C** 25 000 N **D** 50 000 N

117 9702/12/M/J/17/Q7

A rubber ball is dropped onto a table and bounces back up. The table exerts a force *F* on the ball.

Which graph best shows the variation with time *t* of the force *F* for the short time that the ball is in contact with the table?



118 **9702/12/M/J/17/Q8**

A golf ball of mass m is dropped onto a hard surface from a height h_1 and rebounds to a height h_2.

The momentum of the golf ball just as it reaches the surface is different from its momentum just as it leaves the surface.

What is the total change in the momentum of the golf ball between these two instants? (Ignore air resistance.)

A $m\sqrt{2gh_1} - m\sqrt{2gh_2}$

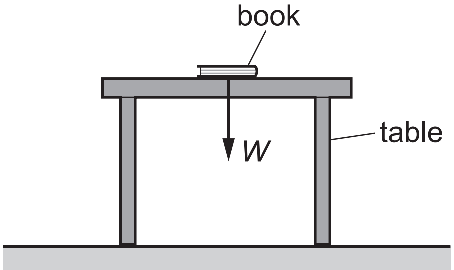
B $m\sqrt{2gh_1} + m\sqrt{2gh_2}$

C $m\sqrt{2g(h_1 - h_2)}$

D $m\sqrt{2g(h_1 + h_2)}$

119 **9702/12/M/J/17/Q9**

A book of weight W is at rest on a table. A student attempts to state Newton's third law of motion by saying that 'action equals reaction'.



If the weight of the book is the 'action' force, what is the 'reaction' force?

A the force W acting downwards on the Earth from the table

B the force W acting upwards on the book from the table

C the force W acting upwards on the Earth from the book

D the force W acting upwards on the table from the floor

120 **9702/12/M/J/17/Q14**

Which quantities are conserved in an inelastic collision?

	kinetic energy	total energy	linear momentum
A	conserved	not conserved	conserved
B	conserved	not conserved	not conserved
C	not conserved	conserved	conserved
D	not conserved	conserved	not conserved

121 **9702/12/M/J/17/Q16**

A stone of mass m falls from rest at the top of a cliff of height h into the sea below. Just before hitting the sea the stone has speed v.

What is the average force of air resistance acting on the stone during its fall?

A mg

B $\dfrac{m(v^2 - 2gh)}{h}$

C $m\left(g - \dfrac{v^2}{2h}\right)$

D $m\left(gh - \dfrac{v^2}{2}\right)$

122 **9702/11/O/N/17/Q8**

The three forces acting on a hot-air balloon that is moving vertically are its weight, the force due to air resistance and the upthrust force.

The hot-air balloon descends vertically at constant speed. The force of air resistance on the balloon is *F*.

Which weight of material must be released from the balloon so that it ascends vertically at the same constant speed?

A F **B** $2F$ **C** $3F$ **D** $4F$

123 **9702/11/O/N/17/Q9**

A car is moving at constant speed in a straight line with the engine providing a driving force equal to the resistive force *F*.

When the engine is switched off, the car is brought to rest in a distance of 100 m by the resistive force.

It may be assumed that *F* is constant during the deceleration.

The process is then repeated for the same car with the same initial speed but with a constant resistive force of $0.800F$.

How far will the car travel while decelerating?

A 120 m **B** 125 m **C** 156 m **D** 250 m

124 **9702/11/O/N/17/Q10**

What is a statement of the principle of conservation of momentum?

A In an elastic collision momentum is constant.

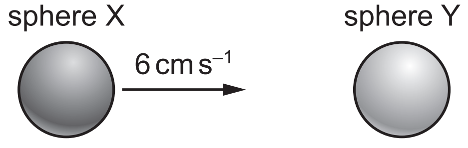
B Momentum is the product of mass and velocity.

C The force acting on a body is proportional to its rate of change of momentum.

D The momentum of an isolated system is constant.

125 **9702/11/O/N/17/Q11**

Two solid spheres form an isolated system. Sphere X moves with speed $6\,\text{cm}\,\text{s}^{-1}$ in a straight line directly towards a stationary sphere Y, as shown.



The spheres have a perfectly elastic collision. After the collision, sphere X moves with speed $2\,\text{cm}\,\text{s}^{-1}$ in the same direction as before the collision.

What is the speed of sphere Y?

A $2\,\text{cm}\,\text{s}^{-1}$ **B** $4\,\text{cm}\,\text{s}^{-1}$ **C** $6\,\text{cm}\,\text{s}^{-1}$ **D** $8\,\text{cm}\,\text{s}^{-1}$

126 **9702/12/O/N/17/Q9**

A slow vehicle and a fast vehicle travel towards each other in a straight line and then collide.

Which outcome is **never** possible, regardless of the masses of the vehicles?

A Both vehicles stop.

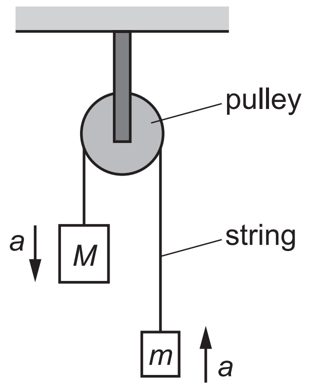
B Only one vehicle stops.

C The fast vehicle's speed increases.

D The slow vehicle's speed increases.

127 **9702/12/O/N/17/Q10**

Two blocks of masses M and m are joined by a thin string which passes over a frictionless pulley, as shown.



The acceleration of free fall is g.

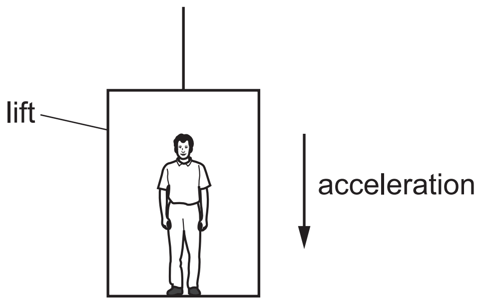
What is the acceleration a of the two blocks?

A $\frac{(M+m)}{(M-m)}g$ **B** $\frac{(M-m)}{(M+m)}g$

C $\frac{M}{m}g$ **D** $\frac{m}{M}g$

128 **9702/13/O/N/17/Q8**

A man stands in a lift that is accelerating vertically downwards, as shown.



Which statement describes the force exerted by the man on the floor?

A It is equal to the weight of the man.

B It is greater than the force exerted by the floor on the man.

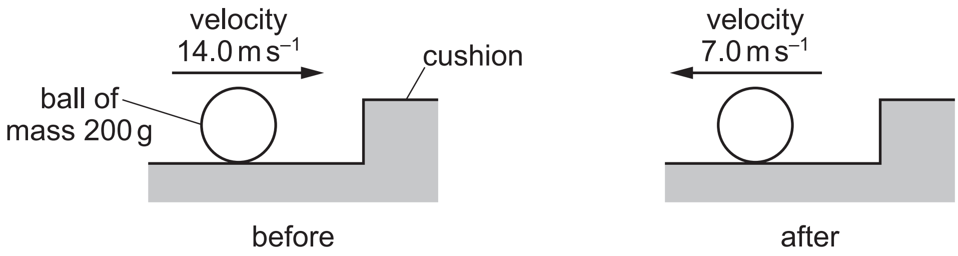
C It is less than the force exerted by the floor on the man.

D It is less than the weight of the man.

129 **9702/13/O/N/17/Q9**

A snooker ball of mass 200 g hits the cushion of a snooker table at right-angles with a speed of $14.0\,\text{m}\,\text{s}^{-1}$.

The ball rebounds with half of its initial speed. The ball is in contact with the cushion for 0.60 s.



What is the average force exerted on the ball by the cushion?

A 2.3 N **B** 7.0 N **C** 2300 N **D** 7000 N

130 **9702/13/O/N/17/Q10**

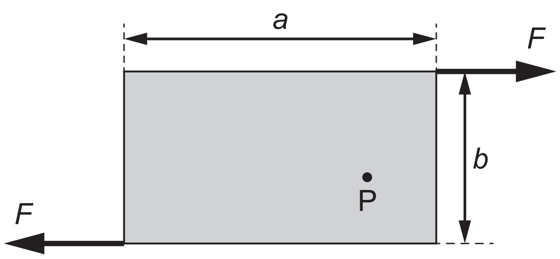
Two railway trucks of masses *m* and 3*m* move towards each other in opposite directions with speeds 2*v* and *v* respectively. These trucks collide and stick together.

What is the speed of the trucks after the collision?

A $\frac{v}{4}$ **B** $\frac{v}{2}$ **C** v **D** $\frac{5v}{4}$

131 **9702/13/O/N/17/Q12**

Two forces, each of magnitude *F*, act along the edges of a rectangular metal plate, as shown.



The plate has length *a* and width *b*. What is the torque about point P?

A *Fa* **B** *Fb* **C** 2*Fa* **D** 2*Fb*

132 **9702/13/O/N/17/Q15**

A ball is thrown vertically upwards. Air resistance is negligible.

Which statement is correct?

A By the principle of conservation of energy, the total energy of the ball is constant throughout its motion.

B By the principle of conservation of momentum, the momentum of the ball is constant throughout its motion.

C The kinetic energy of the ball is greatest at the greatest height attained.

D The potential energy of the ball increases at a constant rate during its ascent.

133 9702/13/O/N/17/Q16

A car of total mass 1560 kg is travelling with a constant speed of $32\,\mathrm{m\,s^{-1}}$. The driving force provided by the car is 680 N. The kinetic energy of the car is 800 kJ and its momentum is 50 000 N s.

Which two items of data could be used to calculate the useful power output of the car?

A driving force and momentum

B kinetic energy and mass

C mass and momentum

D speed and driving force

134 9702/11/M/J/18/Q8

The momentum of a car of mass m increases from p_1 to p_2.

What is the increase in the kinetic energy of the car?

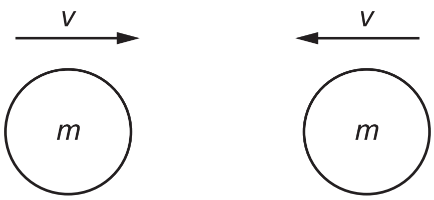
A $\dfrac{(p_2{}^2 - p_1{}^2)}{2m}$ **B** $\dfrac{(p_2 - p_1)^2}{2m}$ **C** $\dfrac{p_2 - p_1}{2m}$ **D** $\dfrac{p_1 - p_2}{2m}$

135 9702/11/M/J/18/Q9

Two similar spheres, each of mass m and travelling with speed v, are moving towards each other.

The spheres have a head-on elastic collision.

Which statement is correct?



A The spheres stick together on impact.

B The total kinetic energy after impact is mv^2.

C The total kinetic energy before impact is zero.

D The total momentum before impact is $2mv$.

136 9702/12/M/J/18/Q8

A tennis ball of mass 55 g is travelling horizontally with a speed of $30\,\mathrm{m\,s^{-1}}$. The ball makes contact with a wall before rebounding in the horizontal direction with a speed of $20\,\mathrm{m\,s^{-1}}$. The ball is in contact with the wall for a time of 5.0×10^{-3} s.

What is the average force exerted on the wall by the ball?

A 110 N **B** 220 N **C** 330 N **D** 550 N

137 9702/12/M/J/18/Q9

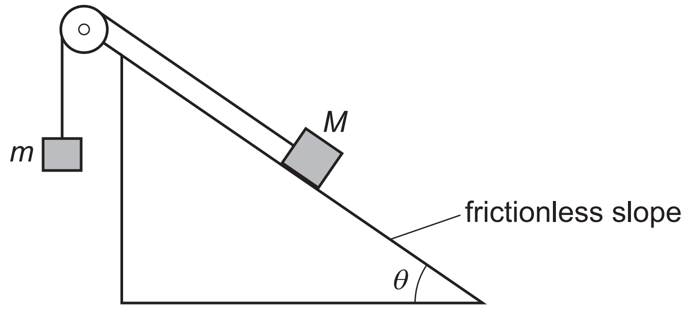
An elastic collision occurs between two bodies X and Y. The mass of body X is m and the mass of body Y is $4m$. Body X travels at speed v before the collision and speed $\frac{3v}{5}$ in the opposite direction after the collision. Body Y is stationary before the collision.

What is the kinetic energy of body Y after the collision?

A $\frac{8}{10}mv^2$ **B** $\frac{34}{50}mv^2$ **C** $\frac{16}{50}mv^2$ **D** $\frac{1}{5}mv^2$

138 9702/13/M/J/18/Q7

Two masses, M and m, are connected by an inextensible string which passes over a frictionless pulley. Mass M rests on a frictionless slope, as shown.



The slope is at an angle θ to the horizontal.

The two masses are initially held stationary and then released. Mass M moves down the slope.

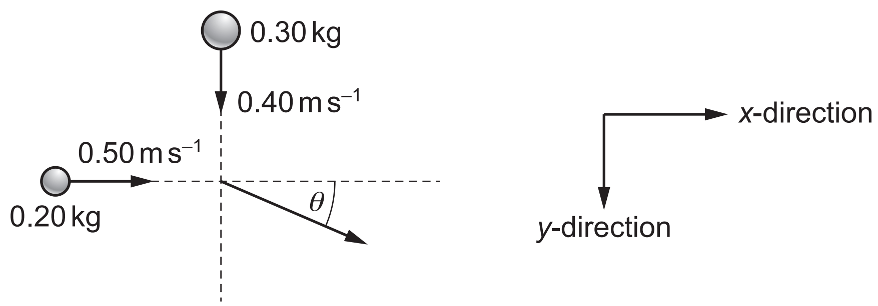
Which expression **must** be correct?

A $\sin\theta < \frac{m}{M}$ **B** $\cos\theta < \frac{m}{M}$ **C** $\sin\theta > \frac{m}{M}$ **D** $\cos\theta > \frac{m}{M}$

139 9702/13/M/J/18/Q9

A ball of mass 0.20 kg, travelling in the x-direction at a speed of $0.50\,\text{m}\,\text{s}^{-1}$, collides with a ball of mass 0.30 kg travelling in the y-direction at a speed of $0.40\,\text{m}\,\text{s}^{-1}$.

The two balls stick together after the collision, travelling at an angle θ to the x-direction.



What is the value of θ?

A 39° **B** 40° **C** 50° **D** 51°

140 **9702/12/F/M/18/Q10**

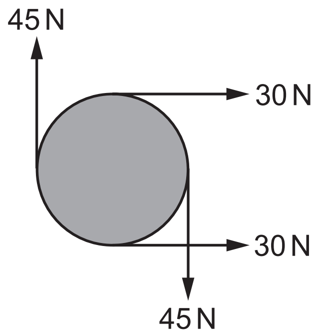
Steel pellets, each with a mass of 0.60 g, fall vertically onto a horizontal plate at a rate of 100 pellets per minute. They strike the plate with a velocity of $5.0\,m\,s^{-1}$ and rebound with a velocity of $4.0\,m\,s^{-1}$.

What is the average force exerted on the plate by the pellets?

A 0.0010 N **B** 0.0054 N **C** 0.0090 N **D** 0.54 N

141 **9702/12/F/M/18/Q11**

The diagram shows four forces applied to a circular object.



Which row describes the resultant force and resultant torque on the object?

	resultant force	resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero

142 **9702/12/F/M/18/Q12**

A charged oil drop is held stationary between two charged parallel plates.

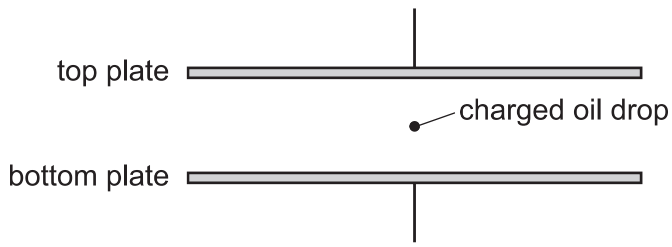
Which forces act on the oil drop?

A both electric and gravitational

B electric only

C gravitational only

D neither electric nor gravitational



143 **9702/11/O/N/18/Q7**

Water is pumped through a hose-pipe at a rate of 90 kg per minute. Water emerges horizontally from the hose-pipe with a speed of $20\,m\,s^{-1}$.

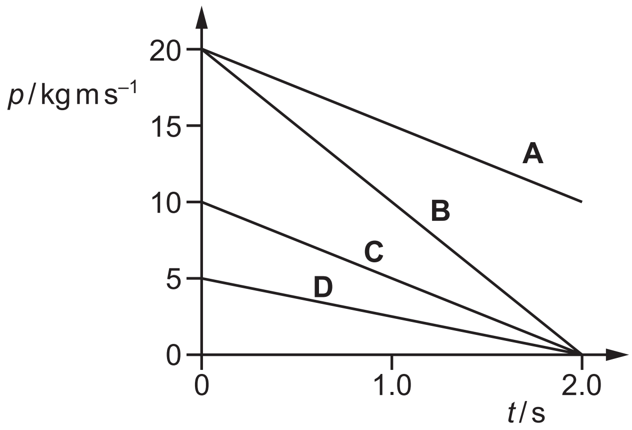
What is the minimum force required from a person holding the hose-pipe to prevent it moving backwards?

A 30 N **B** 270 N **C** 1800 N **D** 108 000 N

144 **9702/12/O/N/18/Q7**

A resultant force of 10 N acts on a body for a time of 2.0 s.

Which graph could show the variation with time *t* of the momentum *p* of the body?



145 **9702/12/O/N/18/Q9**

Two bodies travelling along the same straight line collide in a perfectly elastic collision.

Which statement **must** be correct?

A The initial speed of one body will be the same as the final speed of the other body.

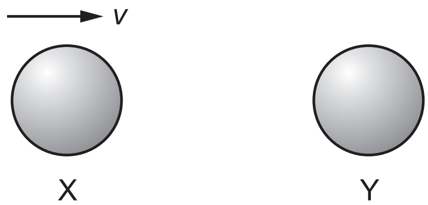
B The relative speed of approach between the two bodies equals their relative speed of separation.

C The total momentum is conserved but the total kinetic energy will be reduced.

D One of the bodies will be stationary at one instant.

146 **9702/12/O/N/18/Q10**

The diagram shows two identical spheres X and Y.



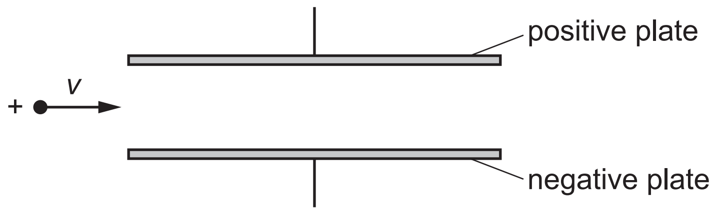
Initially, X moves with speed v directly towards Y. Y is stationary. The spheres collide elastically.

What happens?

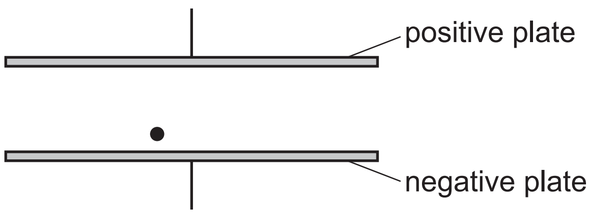
	X	Y
A	moves with speed $\frac{1}{2}v$ to the right	moves with speed $\frac{1}{2}v$ to the right
B	moves with speed v to the left	remains stationary
C	moves with speed $\frac{1}{2}v$ to the left	moves with speed $\frac{1}{2}v$ to the right
D	stops	moves with speed v to the right

147 **9702/12/O/N/18/Q11**

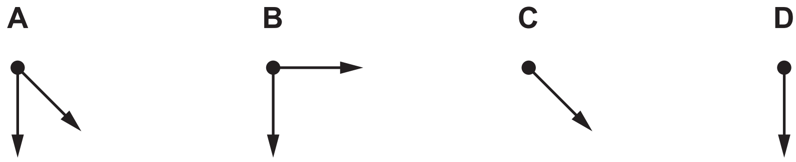
A positively-charged particle of negligible mass, moving at constant velocity v in a vacuum, enters a uniform electric field between two parallel plates, as shown.



A short time later, the particle is at the position shown.



Which diagram represents the force or forces acting on the particle?



148 9702/13/O/N/18/Q7

Two isolated spheres have masses 2.0 kg and 4.0 kg. The spheres collide and then move apart.

During the collision, the 2.0 kg mass has an average acceleration of $8.0\,\mathrm{m\,s^{-2}}$.

What is the average acceleration of the 4.0 kg mass?

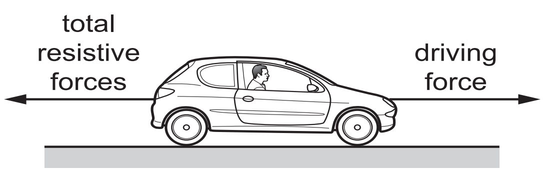
A $2.0\,\mathrm{m\,s^{-2}}$ **B** $4.0\,\mathrm{m\,s^{-2}}$ **C** $8.0\,\mathrm{m\,s^{-2}}$ **D** $16\,\mathrm{m\,s^{-2}}$

149 9702/11/M/J/19/Q9

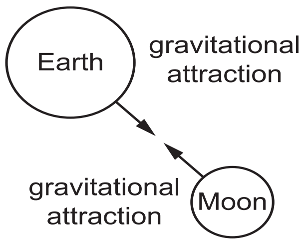
Each diagram illustrates a pair of forces of equal magnitude.

Which diagram gives an example of a pair of forces that is described by Newton's third law of motion?

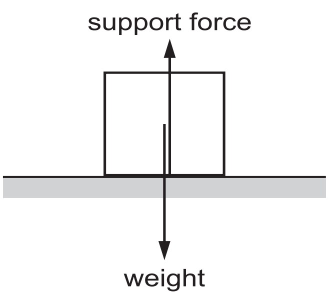
A



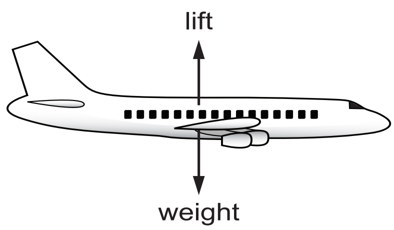
B



C



D

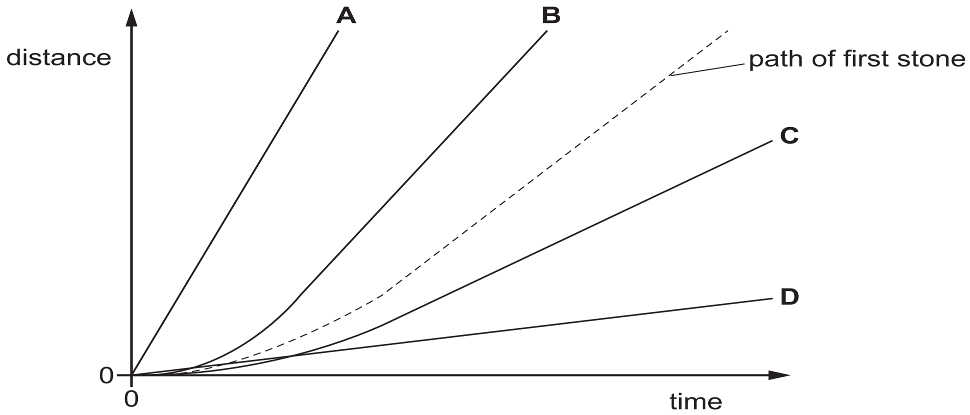


150 9702/11/M/J/19/Q10

A stone is dropped from a tall building. Air resistance is significant. The variation of distance fallen with time is shown by the dashed line.

A second stone with the same dimensions but a smaller mass is dropped from the same building.

Which line represents the motion of the second stone?



151 9702/11/M/J/19/Q11

A helium atom of mass m collides normally with a wall. The atom arrives at the wall with speed v and then rebounds along its original path. Assume that the collision is perfectly elastic.

What is the change in the momentum of the atom during its collision?

A zero **B** $0.5mv$ **C** mv **D** $2mv$

152 9702/12/M/J/19/Q9

What describes the mass of an object?

A the force the object experiences due to gravity

B the momentum of the object before a collision

C the resistance of the object to changes in motion

D the weight of the object as measured by a balance

153 9702/12/M/J/19/Q10

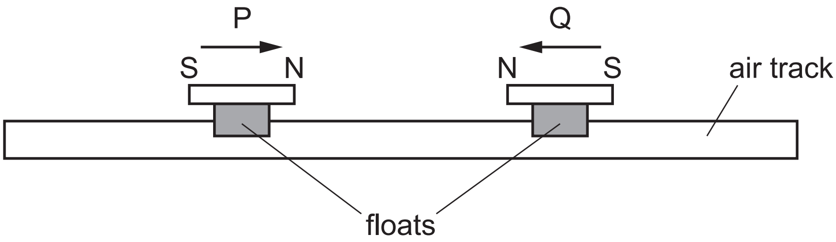
A car has mass m. A person needs to push the car with force F in order to give the car acceleration a. The person needs to push the car with force $2F$ in order to give the car acceleration $3a$.

Which expression gives the constant resistive force opposing the motion of the car?

A ma **B** $2ma$ **C** $3ma$ **D** $4ma$

154 **9702/12/M/J/19/Q11**

Two bar magnets P and Q are mounted on floats which can slide without friction along an air track.



The two magnets slide towards each other along the air track and interact, without making contact.

The relative speed of approach of the magnets is equal to their relative speed of separation.

Which statement about P and Q must be correct?

A During the interaction between P and Q some of the total kinetic energy is lost.

B During the interaction between P and Q some of the total momentum is lost.

C The momentum of Q after the interaction is equal to the momentum of P before the interaction.

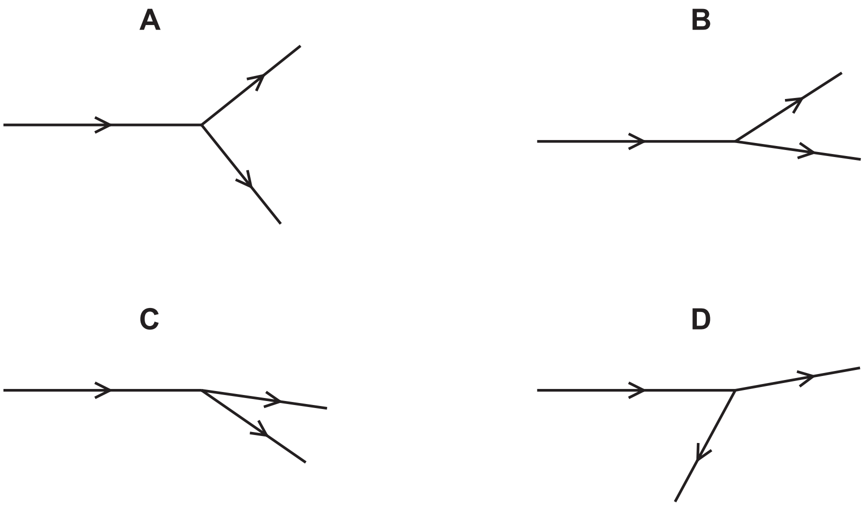
D The values of (kinetic energy of P + kinetic energy of Q) before and after the interaction are equal.

155 **9702/13/M/J/19/Q9**

A nucleus collides with a stationary nucleus in a vacuum. The diagrams show the paths of the nuclei before and after the collision.

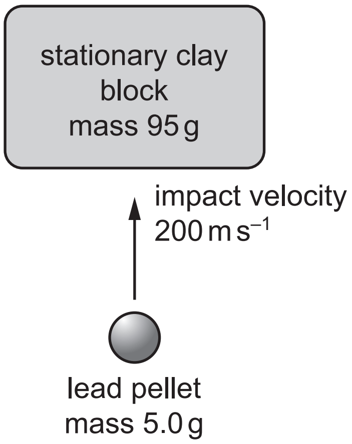
No other particles are involved in the collision.

Which diagram is **not** possible?



156 **9702/13/M/J/19/Q17**

A lead pellet is shot vertically upwards into a clay block that is stationary at the moment of impact but is able to rise freely after impact.



The mass of the pellet is 5.0 g and the mass of the clay block is 95 g.

The pellet hits the block with an initial vertical velocity of $200\,m\,s^{-1}$. It embeds itself in the block and does not emerge.

How high above its initial position will the block rise?

A 5.1 m **B** 5.6 m **C** 10 m **D** 100 m

157 **9702/11/O/N/19/Q7**

Which statement follows directly from Newton's first law?

A A body remains at constant velocity unless acted upon by a resultant force.

B A satellite in circular motion about the Earth has a constant velocity.

C A water drop leaving a spinning umbrella travels at a constant velocity.

D The force acting on an object is equal to its change in momentum.

158 **9702/11/O/N/19/Q8**

A resultant force causes an object to accelerate.

What is equal to the resultant force?

A the acceleration of the object per unit mass

B the change in kinetic energy of the object per unit time

C the change in momentum of the object per unit time

D the change in velocity of the object per unit time

159 **9702/11/O/N/19/Q9**

A skydiver falls from an aircraft that is moving horizontally.

The vertical component of the velocity of the skydiver is *v*.

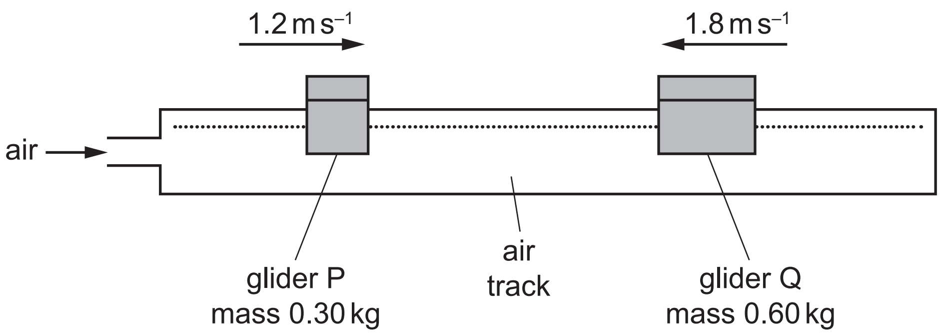
The vertical component of the acceleration of the skydiver is *a*.

Which row describes *v* and *a* during the first few seconds after the skydiver leaves the aircraft?

	v	*a*
A	constant	constant
B	constant	decreasing
C	increasing	constant
D	increasing	decreasing

160 **9702/11/O/N/19/Q10**

Two gliders are travelling towards each other on a horizontal air track. Glider P has mass 0.30 kg and is moving with a constant speed of $1.2\,\mathrm{m\,s^{-1}}$. Glider Q has mass 0.60 kg and is moving with a constant speed of $1.8\,\mathrm{m\,s^{-1}}$.



The gliders have a perfectly elastic collision.

What are the speeds of the two gliders after the collision?

	speed of P / $\mathrm{m\,s^{-1}}$	speed of Q / $\mathrm{m\,s^{-1}}$
A	1.2	0.6
B	2.0	1.4
C	2.8	0.2
D	3.6	0.6

161 **9702/12/O/N/19/Q7**

A snooker ball has a mass of 200 g. It hits the cushion of a snooker table and rebounds along its original path.

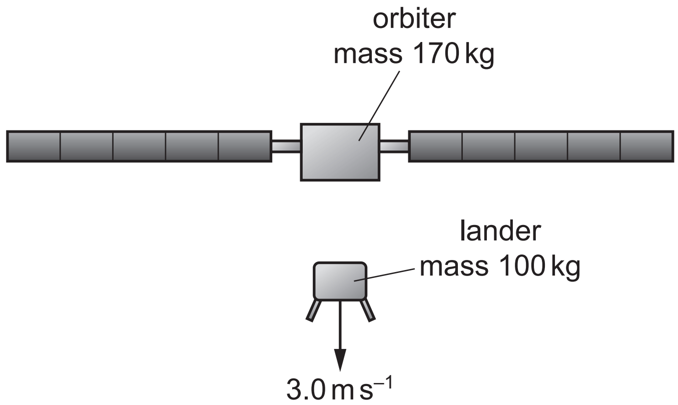
The ball arrives at the cushion with a speed of $14.0\,\mathrm{m\,s^{-1}}$ and then leaves it with a speed of $7.0\,\mathrm{m\,s^{-1}}$. The ball and the cushion are in contact for a time of 0.60 s.

What is the average force exerted on the ball by the cushion?

A 1.4 N **B** 2.3 N **C** 4.2 N **D** 7.0 N

162 **9702/12/O/N/19/Q9**

The space probe Rosetta was designed to investigate a comet. The probe consisted of an orbiter and a lander. The orbiter had a mass of 170 kg and the lander had a mass of 100 kg. When the two parts separated, the lander was pushed towards the surface of the comet so that its change in velocity towards the comet was $3.0\,\mathrm{m\,s^{-1}}$.



Assume that the orbiter and lander were an isolated system.

The orbiter moved away from the comet during the separation.

What was the change in the speed of the orbiter?

A $1.8\,\mathrm{m\,s^{-1}}$ **B** $2.3\,\mathrm{m\,s^{-1}}$ **C** $3.0\,\mathrm{m\,s^{-1}}$ **D** $5.1\,\mathrm{m\,s^{-1}}$

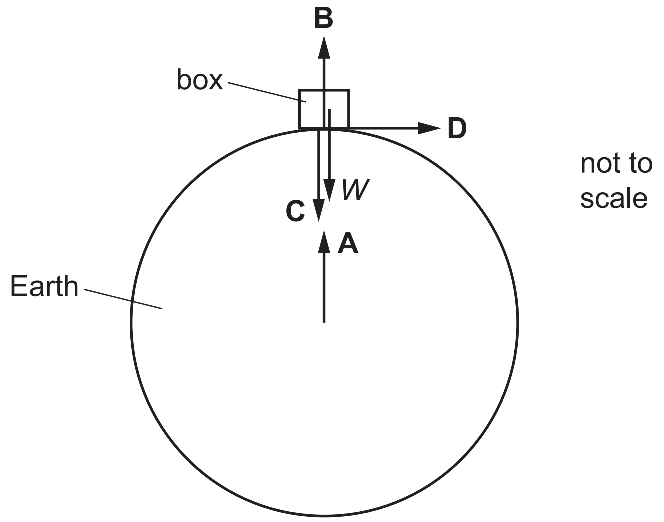
-

163 **9702/13/O/N/19/Q7**

A box rests on the Earth, as shown.

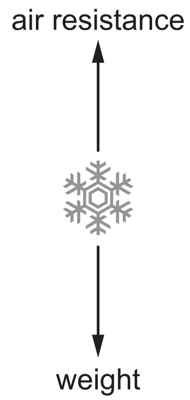
Newton's third law describes how forces of the same type act in pairs. One of the forces of a pair is the weight *W* of the box.

Which arrow represents the other force of this pair?



164 **9702/13/O/N/19/Q8**

A snowflake is falling from the sky on a still day. Its weight acts vertically downwards and air resistance acts vertically upwards. As the snowflake falls, air resistance increases until it is equal to the weight and there is no resultant force acting on the snowflake.



When the forces become equal, which statement is correct?

A The snowflake accelerates.

B The snowflake decelerates.

C The snowflake is stationary.

D The snowflake moves at a constant velocity.

165 9702/13/O/N/19/Q9

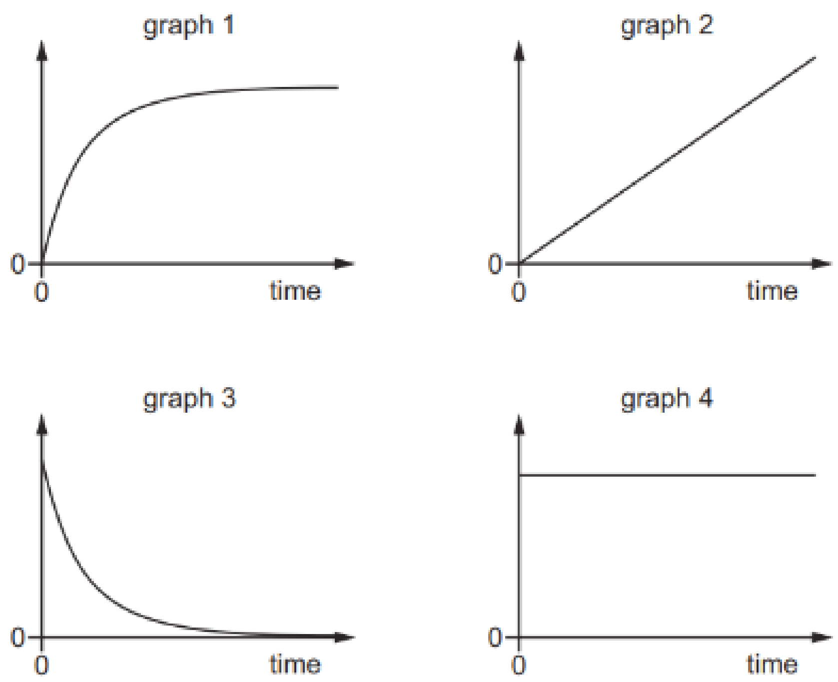
Two objects X and Y in an isolated system undergo a perfectly elastic collision. The velocities of the objects before and after the collision are shown.

What is the speed v of Y after the collision?

A $2.0\,m\,s^{-1}$ **B** $18\,m\,s^{-1}$ **C** $22\,m\,s^{-1}$ **D** $24\,m\,s^{-1}$

166 9702/12/F/M/20/Q9

The diagram shows graphs of various quantities plotted against time for an object dropped from a stationary balloon high in the atmosphere.



Which statement could be correct?

A Graph 1 is acceleration against time and graph 3 is resultant force against time.

B Graph 1 is acceleration against time and graph 4 is resultant force against time.

C Graph 3 is acceleration against time and graph 1 is velocity against time.

D Graph 3 is acceleration against time and graph 2 is velocity against time.

167. 9702/12/F/M/20/Q8

A person of mass 60 kg stands on accurate bathroom scales, placed on the floor of an elevator (lift) which operates in a tall building.

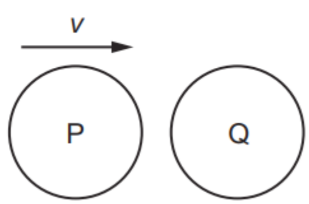
At a certain instant the bathroom scales read 58 kg.

Which row could give the person's direction of movement and type of motion?

	direction	motion
A	downwards	constant speed
B	downwards	slowing down
C	upwards	constant speed
D	upwards	slowing down

168. 9702/12/F/M/20/Q10

The diagram shows a particle P, travelling at speed *v*, about to collide with a stationary particle Q of the same mass. The collision is perfectly elastic.



Which statement describes the motion of P and of Q immediately after the collision?

A P and Q both travel in the same direction with speed $\frac{1}{2}v$.

B P comes to rest and Q acquires speed v.

C P rebounds with speed $\frac{1}{2}v$ and Q acquires speed $\frac{1}{2}v$.

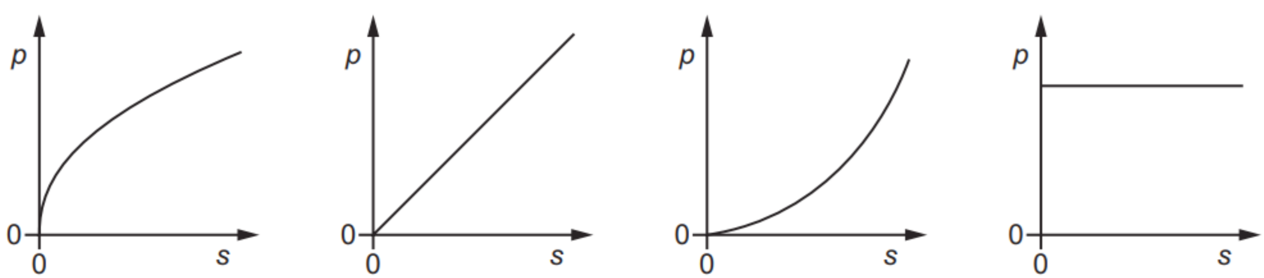
D P rebounds with speed v and Q remains stationary.

169. 9702/11/M/J/20/Q8

A car accelerates from rest in a straight line with constant acceleration.

Which graph best represents the variation of the momentum *p* of the car with the distance *s* travelled by the car?

A **B** **C** **D**



170. 9702/11/M/J/20/Q10

A stationary toy gun fires a bullet.

Which statement about the bullet and the gun, immediately after firing, is **not** correct?

A The force exerted on the bullet by the gun has the same magnitude as the force exerted on the gun by the bullet.

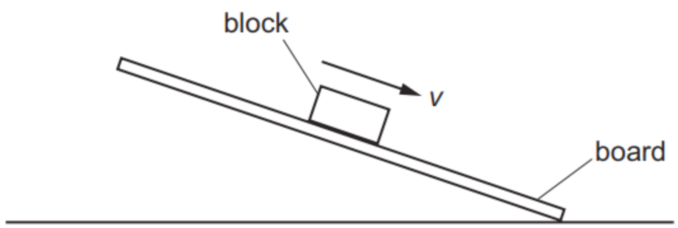
B The force exerted on the bullet by the gun is in the opposite direction to the force exerted on the gun by the bullet.

C The gun and the bullet have the same magnitude of momentum.

D The kinetic energy of the gun must equal the kinetic energy of the bullet.

171. 9702/11/M/J/20/Q11

A wooden block rests on the rough surface of a board. One end of the board is then raised until the block slides down the board at constant velocity *v*.



What describes the forces acting on the block when it is sliding with constant velocity?

	frictional force on block	resultant force on block
A	down the board	down the board
B	down the board	zero
C	up the board	down the board
D	up the board	zero

172. 9702/12/M/J/20/Q8

An astronaut has a weight of 660 N when she is standing on the Earth's surface.

The acceleration of free fall on the surface of Mars is $3.71\,\mathrm{m\,s^{-2}}$.

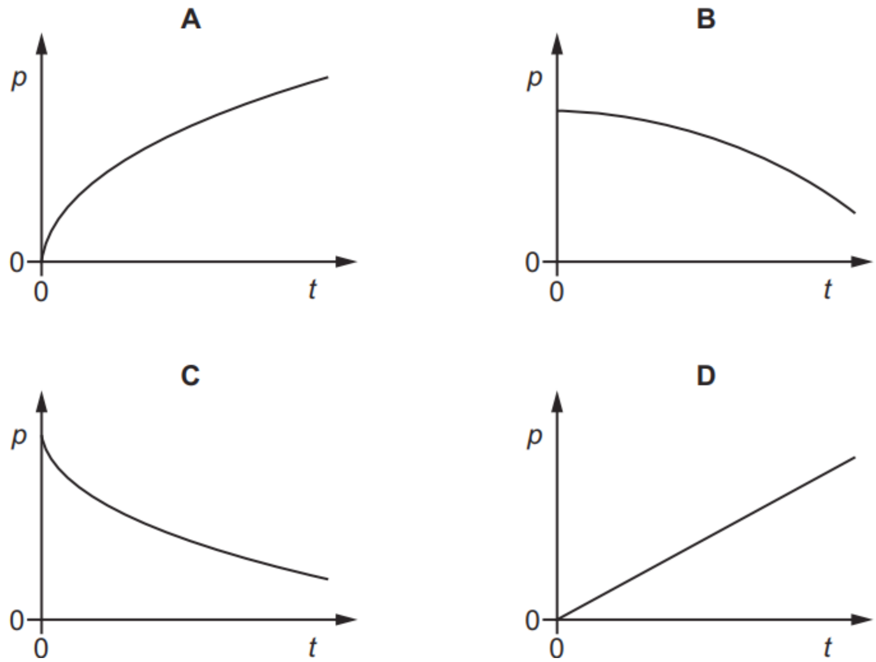
What would be the weight of the astronaut if she stood on the surface of Mars?

A 67.3 N **B** 178 N **C** 250 N **D** 660 N

173. 9702/12/M/J/20/Q7

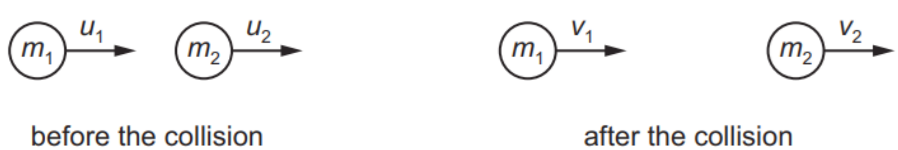
The resultant force acting on an object is slowly increased.

Which graph could show the variation with time t of the momentum p of the object?



174. 9702/12/M/J/20/Q9

A mass m_1 travelling with speed u_1 collides with a mass m_2 travelling with speed u_2 in the same direction. After the collision, mass m_1 has speed v_1 and mass m_2 has speed v_2 in the same direction. The collision is perfectly elastic.



Which equation is **not** correct?

A $m_1u_1^2 - m_1v_1^2 = m_2v_2^2 - m_2u_2^2$

B $v_2 + u_2 = v_1 + u_1$

C $m_1(u_1 - v_1) = m_2(v_2 - u_2)$

D $m_1(u_1 - v_1)^2 = m_2(u_2 - v_2)^2$

175. 9702/13/M/J/20/Q8

A ball of mass m travels vertically downwards and then hits a horizontal floor at speed u.

It rebounds vertically upwards with speed v.

The collision lasts a time Δt.

What is the average resultant force exerted on the ball during the collision?

A $\dfrac{mv - mu}{\Delta t}$ downwards

B $\dfrac{mv - mu}{\Delta t}$ upwards

C $\dfrac{mv + mu}{\Delta t}$ downwards

D $\dfrac{mv + mu}{\Delta t}$ upwards

176. 9702/13/M/J/20/Q10

A ball of mass m, moving at a velocity v, collides with a stationary ball of mass $2m$.

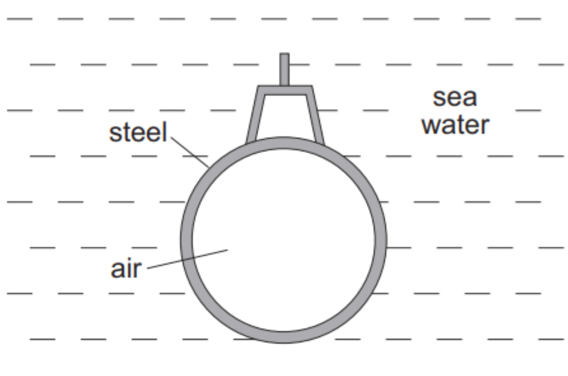
The two balls stick together.

Which fraction of the initial kinetic energy is lost on impact?

A $\frac{1}{9}$ **B** $\frac{1}{3}$ **C** $\frac{2}{3}$ **D** $\frac{8}{9}$

177. 9702/13/M/J/20/Q11

A submarine is in equilibrium in a fully submerged position.



What causes the upthrust on the submarine?

A The air in the submarine is less dense than sea water.

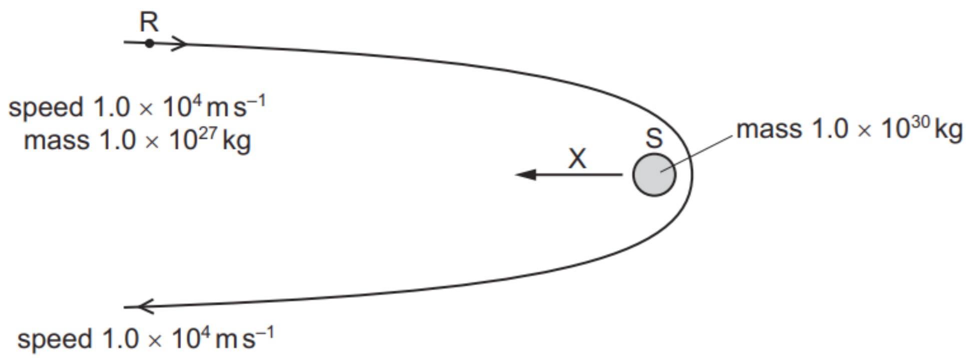
B There is a difference in water pressure acting on the top and on the bottom of the submarine.

C The sea water exerts a greater upward force on the submarine than the weight of the steel.

D The submarine displaces its own volume of sea water.

178. 9702/11/O/N/20/Q9

A rock R of mass 1.0×10^{27} kg is a large distance from a star S and is travelling at a speed of $1.0 \times 10^{4}\,m\,s^{-1}$. The star has mass 1.0×10^{30} kg. The rock travels around the star on the path shown so that it reverses its direction of motion and, when finally again a large distance from the star, has the same speed as initially.



Which statement is correct?

A The change in the momentum of S is in the direction of arrow X.

B The change in the velocity of S is approximately $20\,m\,s^{-1}$.

C The magnitude of the change of momentum of R is 10^{3} times greater than the magnitude of the change of momentum of S.

D The momentum of R does not change.

179. 9702/11/O/N/20/Q10

The diagram shows the masses and velocities of two trolleys that are about to collide.

After the impact they move off together.

What is the kinetic energy lost in the collision?

A 4 J **B** 6 J **C** 12 J **D** 14 J

180. 9702/12/O/N/20/Q8

What is **not** a statement of one of Newton's laws of motion?

A If body X exerts a force on body Y, body Y exerts an equal and opposite force on body X.

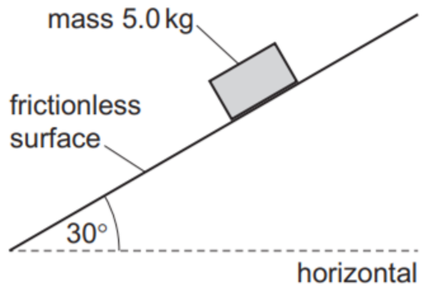
B If no resultant force acts on a body it has constant velocity.

C The rate of change of momentum of a body is proportional to the resultant force acting on it and takes place in the direction of the force.

D The total momentum of a system of interacting bodies is constant if there is no external force.

181. 9702/12/O/N/20/Q7

A mass of 5.0 kg is released from rest on a frictionless surface inclined at 30° to the horizontal. Air resistance is negligible.



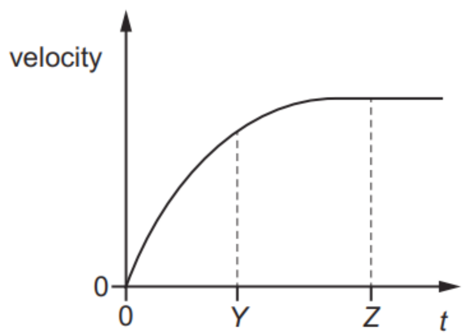
How far does the mass travel in a time of 0.80 s?

A 1.6 m **B** 2.0 m **C** 2.7 m **D** 3.1 m

182. 9702/12/O/N/20/Q9

An object falls from a tall building.

The graph shows how the velocity of the object changes with time *t*.



The acceleration of free fall is *g*.

What describes the acceleration of the object at times $t = Y$ and $t = Z$?

	acceleration at $t = Y$	acceleration at $t = Z$
A	decreasing	g
B	decreasing	0
C	constant	g
D	constant	0

183. 9702/12/O/N/20/Q10

Two balls, one of mass $2m$ and one of mass m, collide.

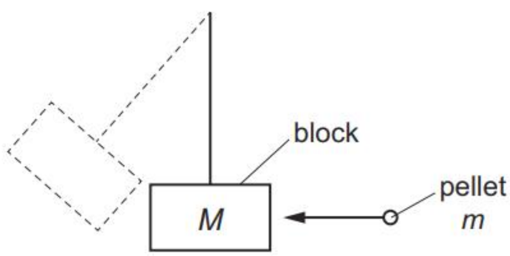
The diagrams show the initial and final velocities of the balls.

Which collision is **not** elastic?

	before collision	after collision
A	$2m$: $4.0\,\text{m s}^{-1}$ → ; m: $1.0\,\text{m s}^{-1}$ →	$2m$: $2.0\,\text{m s}^{-1}$ → ; m: $5.0\,\text{m s}^{-1}$ →
B	$2m$: $6.0\,\text{m s}^{-1}$ → ; m: $3.0\,\text{m s}^{-1}$ →	$2m$: $4.0\,\text{m s}^{-1}$ → ; m: $7.0\,\text{m s}^{-1}$ →
C	$2m$: $8.0\,\text{m s}^{-1}$ → ; m: $2.0\,\text{m s}^{-1}$ →	$2m$: $5.0\,\text{m s}^{-1}$ → ; m: $8.0\,\text{m s}^{-1}$ →
D	$2m$: $10.0\,\text{m s}^{-1}$ → ; m: $4.0\,\text{m s}^{-1}$ →	$2m$: $6.0\,\text{m s}^{-1}$ → ; m: $12.0\,\text{m s}^{-1}$ →

184. 9702/12/O/N/20/Q11

The diagram shows a 'ballistic pendulum'.



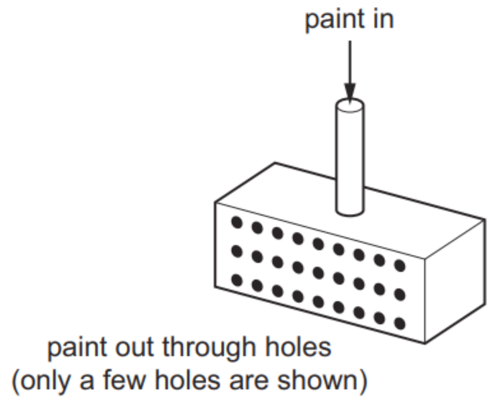
A pellet of mass m travelling at a speed u hits a stationary block of mass M. The pellet becomes embedded in the block and causes the block to move at a speed v immediately after the impact.

When a pellet of mass $2m$, travelling at a speed $2u$, hits a block of mass $2M$, what is the speed of the block immediately after the impact? (Neglect the small increase in the mass of the block as the pellet's mass is added during the collision.)

A v **B** $v\sqrt{2}$ **C** $2v$ **D** $4v$

185. 9702/13/O/N/20/Q8

A device for spraying paint consists of a box with its axes horizontal and vertical. One of its vertical faces contains small holes. Paint is fed into the box under pressure via a vertical tube and exits through the holes as fine streams moving horizontally.



The paint is ejected at a speed of $2.5\,m\,s^{-1}$ through 400 holes, each of area $0.4\,mm^2$. The density of the paint is $900\,kg\,m^{-3}$.

What is the horizontal force required to hold the device stationary as it ejects the paint?

A 0.36 N **B** 0.90 N **C** 2.3 N **D** 900 N

186. 9702/13/O/N/20/Q9

A party balloon is filled with air and held stationary at a height of several metres above the ground. The balloon is then dropped in still air.

Which statement describes the motion of the balloon from the moment of release until just before it hits the floor?

A The balloon decelerates continuously.

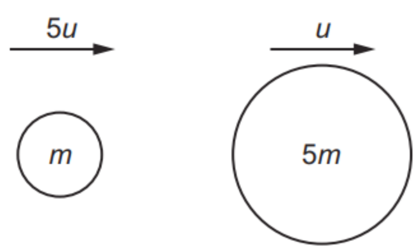
B The balloon falls at a constant speed and then decelerates.

C The balloon falls at a constant speed.

D The balloon initially accelerates and then reaches a constant speed.

187. 9702/13/O/N/20/Q10

An object of mass *m* travelling with speed 5*u* collides with, and sticks to, an object of mass 5*m* travelling in the same direction with speed *u*.



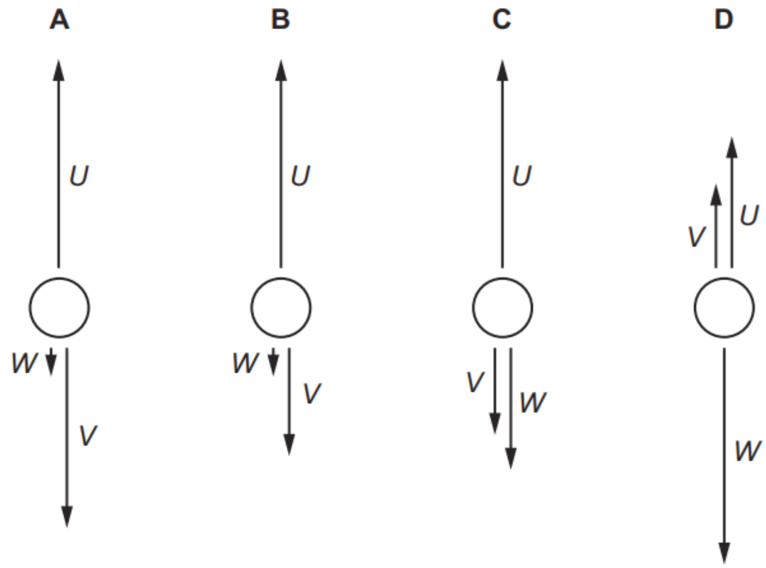
What is the speed with which the two objects travel together in the original direction?

A $\frac{3}{10}u$ **B** u **C** $\frac{6}{5}u$ **D** $\frac{10}{6}u$

188. 9702/13/O/N/20/Q11

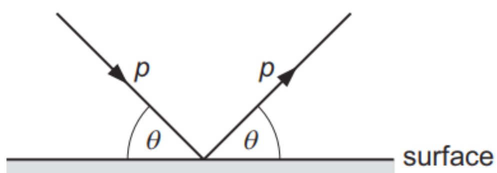
An air bubble is rising through a liquid at a constant speed. The forces on it are the upthrust U, the viscous drag V and its weight W.

Which diagram shows the directions and relative sizes of the forces?



189. 9702/12/F/M/21/Q8

A ball strikes a horizontal surface with momentum p at an angle θ to the surface, as shown.



The ball rebounds with the same magnitude of momentum at an angle θ to the surface.

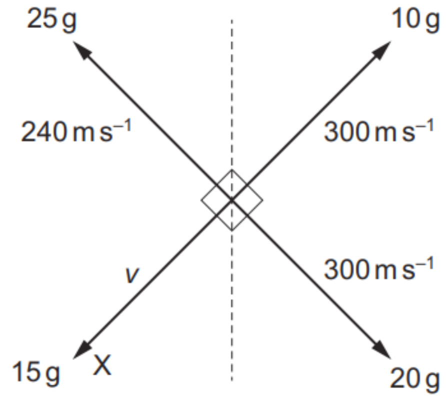
The ball is in contact with the surface for time t.

What is the magnitude of the average resultant force acting on the ball during the collision?

A zero **B** $\dfrac{2p}{t}$ **C** $\dfrac{2p\cos\theta}{t}$ **D** $\dfrac{2p\sin\theta}{t}$

190. 9702/12/F/M/21/Q10

A stationary firework explodes into four fragments which travel in different directions in a horizontal plane. The diagram shows the velocity and mass of each fragment.



NOT TO SCALE

What is the speed v of fragment X?

A $200\,\mathrm{m\,s^{-1}}$ **B** $240\,\mathrm{m\,s^{-1}}$ **C** $300\,\mathrm{m\,s^{-1}}$ **D** $360\,\mathrm{m\,s^{-1}}$

191. 9702/11/M/J/21/Q8

A rocket is fired from the Earth into space.

Newton's third law of motion describes how forces act in pairs. One of the forces of a pair is the weight of the rocket.

What is the other force of this pair?

A air resistance

B force of the exhaust gases on the rocket

C force of the rocket on the exhaust gases

D gravitational force of the rocket on the Earth

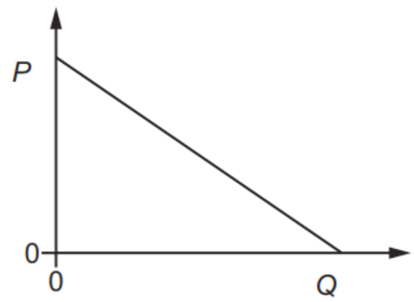
192. 9702/11/M/J/21/Q10

Which quantities are conserved in an inelastic collision?

	kinetic energy	total energy	linear momentum
A	conserved	not conserved	conserved
B	conserved	not conserved	not conserved
C	not conserved	conserved	conserved
D	not conserved	conserved	not conserved

193. 9702/11/M/J/21/Q9

The graph shows how quantity *P* varies with quantity *Q* for a body falling vertically downwards in a uniform gravitational field with air resistance.

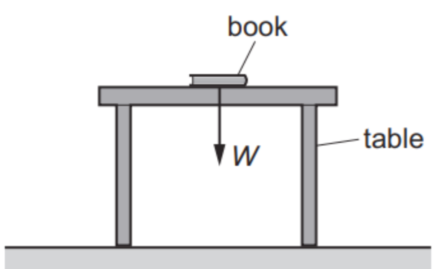


Which pair of quantities could be represented by *P* and *Q*?

	P	*Q*
A	acceleration	force of air resistance
B	acceleration	time
C	velocity	force of air resistance
D	velocity	time

194. 9702/12/M/J/21/Q8

A book of weight *W* is at rest on a table. A student attempts to state Newton's third law of motion by saying that 'action equals reaction'.



If the weight of the book is the 'action' force, what is the 'reaction' force?

A the force *W* acting downwards on the Earth from the table

B the force *W* acting upwards on the book from the table

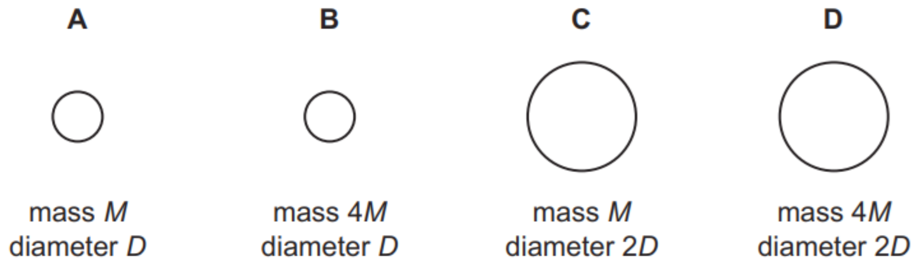
C the force *W* acting upwards on the Earth from the book

D the force *W* acting upwards on the table from the floor

195. 9702/12/M/J/21/Q9

Four balls are dropped at the same time from the top of a very tall tower. There is no wind blowing.

Which ball hits the ground first?

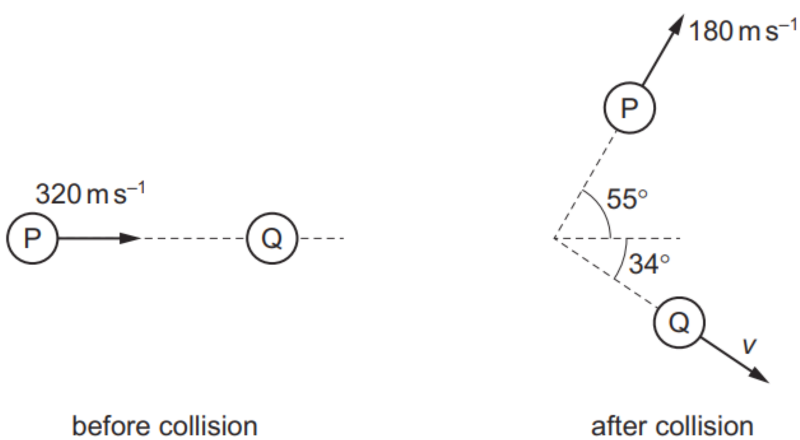


196. 9702/12/M/J/21/Q10

A nitrogen molecule P travelling at a speed of $320\,\text{m}\,\text{s}^{-1}$ in a vacuum collides with a stationary nitrogen molecule Q.

After the collision, P travels at a velocity of $180\,\text{m}\,\text{s}^{-1}$ at an angle of 55° to its original path.

Q travels in a direction at an angle of 34° to the initial path of P.



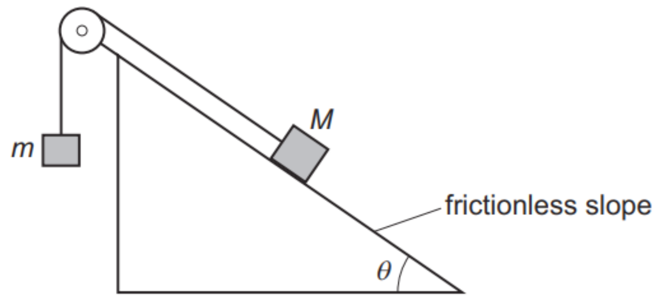
Assume that there are no external forces acting on the molecules.

What is the magnitude v of the velocity of Q after the collision?

A $120\,\text{m}\,\text{s}^{-1}$ **B** $140\,\text{m}\,\text{s}^{-1}$ **C** $180\,\text{m}\,\text{s}^{-1}$ **D** $260\,\text{m}\,\text{s}^{-1}$

197. 9702/13/M/J/21/Q8

Two masses, M and m, are connected by an inextensible string which passes over a frictionless pulley. Mass M rests on a frictionless slope, as shown.



The slope is at an angle θ to the horizontal.

The two masses are initially held stationary and then released. Mass M accelerates down the slope.

Which expression **must** be correct?

A $\sin\theta < \frac{m}{M}$ **B** $\cos\theta < \frac{m}{M}$ **C** $\sin\theta > \frac{m}{M}$ **D** $\cos\theta > \frac{m}{M}$

198. 9702/13/M/J/21/Q9

The weights and masses of four different objects on the surfaces of four different planets are shown.

Which planet has the lowest value of acceleration of free fall at its surface?

	weight	mass
A	40 mN	6.0 g
B	3.0 N	500 g
C	10 N	1 kg
D	2.6 kN	750 kg

199. 9702/13/M/J/21/Q10

A rock in deep space is travelling towards a distant star and collides with a stationary spacecraft.

What is **not** a possible outcome of the collision?

A The rock becomes stationary and the spacecraft moves towards the star.

B The rock moves away from the star and so does the spacecraft.

C The rock moves away from the star and the spacecraft moves towards the star.

D The rock moves towards the star and so does the spacecraft.

200. 9702/13/M/J/21/Q11

A steel ball is falling at a constant (terminal) speed in still air. The forces acting on the ball are upthrust, viscous drag and weight.

What is the order of increasing magnitude of these three forces?

A upthrust → viscous drag → weight

B viscous drag → upthrust → weight

C viscous drag → weight → upthrust

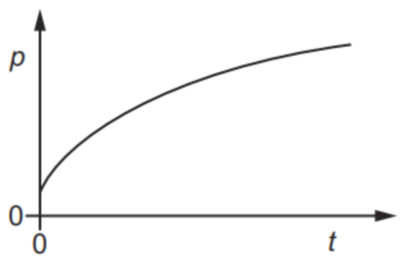
D weight → upthrust → viscous drag

201. 9702/11/O/N/21/Q8

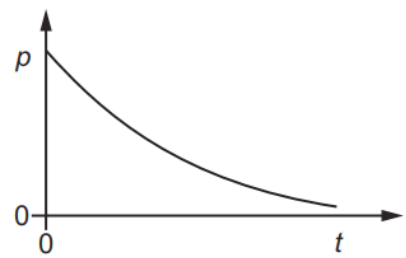
A constant resultant force acts on an object in the direction of the object's velocity.

Which graph could show the variation with time t of the momentum p of the object?

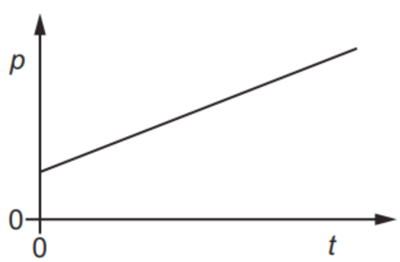
A



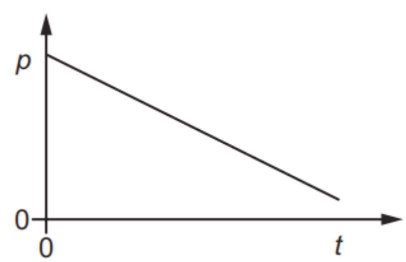
B



C



D



202. 9702/11/O/N/21/Q9

Which statement **must** be true for an object in a gravitational field?

A If the object has mass then the field causes it to accelerate.

B If the object has mass then the field causes it to have weight.

C If the object has weight then the field causes it to accelerate.

D If the object has weight then the field causes it to have mass.

203. 9702/11/O/N/21/Q10

A ball of mass 0.16 kg is travelling horizontally at a speed of $20\,m\,s^{-1}$.

It collides with a wall and rebounds with a speed of $15\,m\,s^{-1}$ along its original path. The ball is in contact with the wall for a time of 1.0 ms.

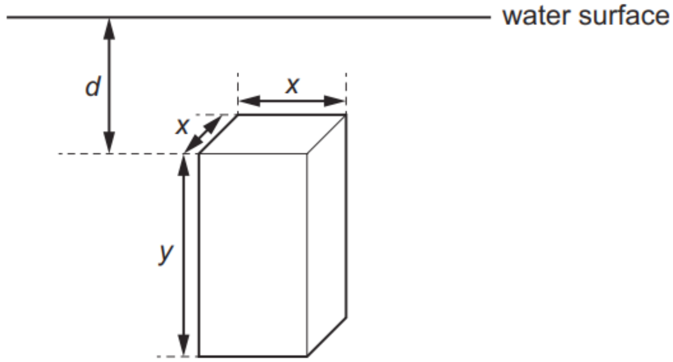
What is the average force exerted by the wall on the ball?

A 800 N **B** 2400 N **C** 3200 N **D** 5600 N

204. 9702/11/O/N/21/Q11

A uniform solid block is fully submerged in a tank of water.

The dimensions of the block are *x* and *y*, as shown.



The block is held vertically in the position shown. The density of the block is the same as the density of the water.

If the block is always held at the same depth *d* below the surface of the water, which single change would **increase** the magnitude of the upthrust force on the block?

A decrease the density of the block

B hold the block horizontally

C increase dimension *y*

D increase the density of the block

205. 9702/12/O/N/21/Q8

What is meant by the mass of an object?

A the property of the object that resists a change in motion

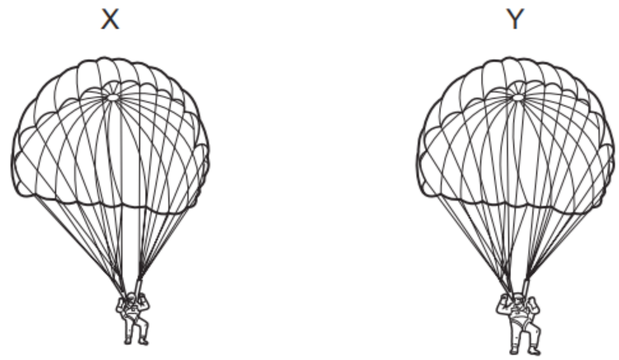
B the pull of the Earth on the object

C the total number of atoms in the object

D the weight of the object

206. 9702/12/O/N/21/Q9

The diagram shows two parachutists, X and Y, moving vertically downwards.



The total mass of parachutist Y and his parachute is twice the total mass of parachutist X and his parachute. At this moment, the air resistance on parachute Y is twice the air resistance on parachute X. Neither parachutist has reached his constant (terminal) velocity.

Which statement describes the acceleration of Y compared with the acceleration of X?

A The acceleration of Y is half the acceleration of X.

B The acceleration of Y is the same as the acceleration of X.

C The acceleration of Y is more than the acceleration of X, but less than twice the value.

D The acceleration of Y is twice the acceleration of X.

207. 9702/12/O/N/21/Q10

The table shows four different collisions between two blocks, each of mass 0.50 kg.

Which collision is perfectly elastic?

	before collision		after collision	
A	$4.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg	$0.0\,\text{m s}^{-1}$ 0.50 kg	$2.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg 0.50 kg	
B	$2.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg	$\leftarrow 2.0\,\text{m s}^{-1}$ 0.50 kg	$0.0\,\text{m s}^{-1}$ 0.50 kg 0.50 kg	
C	$2.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg	$\leftarrow 1.0\,\text{m s}^{-1}$ 0.50 kg	$\leftarrow 2.0\,\text{m s}^{-1}$ 0.50 kg	$3.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg
D	$4.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg	$1.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg	$1.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg	$4.0\,\text{m s}^{-1} \rightarrow$ 0.50 kg

208. 9702/12/O/N/21/Q11

A cylindrical block of wood has cross-sectional area A and weight W. It is totally immersed in water with its axis vertical. The block experiences pressures p_t and p_b at its top and bottom surfaces respectively.

What is the upthrust on the block?

A $(p_b - p_t)$

B $(p_b - p_t)A$

C $(p_b - p_t)A - W$

D $(p_b - p_t)A + W$

209. 9702/13/O/N/21/Q8

Water is pumped through a hose-pipe at a rate of 90 kg per minute. Water emerges horizontally from the hose-pipe with a speed of $20\,m\,s^{-1}$.

What is the minimum force required from a person holding the hose-pipe to prevent it moving backwards?

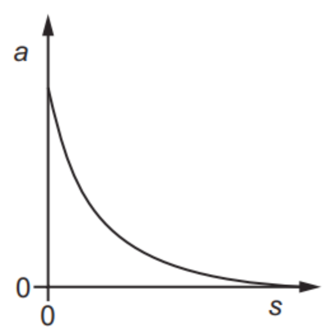
A 30 N **B** 270 N **C** 1800 N **D** 110 000 N

210. 9702/13/O/N/21/Q9

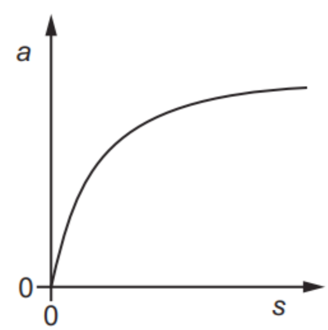
A skydiver leaves a stationary balloon and falls vertically through a long distance.

Which graph best represents the variation of the acceleration a of the skydiver with the distance s travelled as she falls through the air?

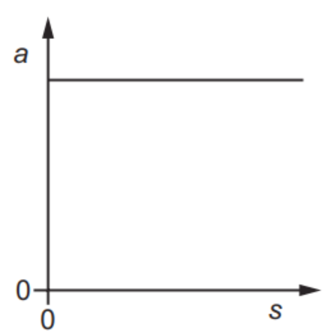
A



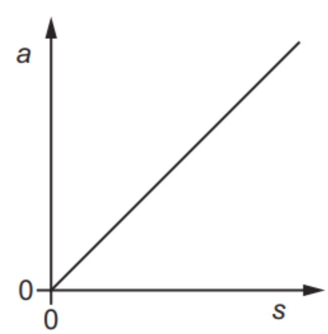
B



C



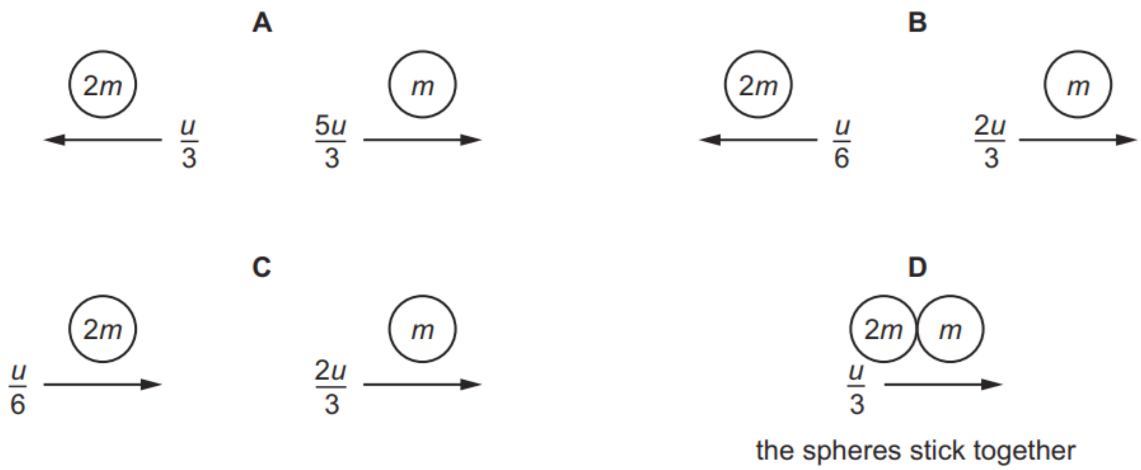
D



211. 9702/13/O/N/21/Q10

The diagram shows two spheres approaching each other head-on. Each sphere has speed u. One sphere has mass $2m$ and the other has mass m.

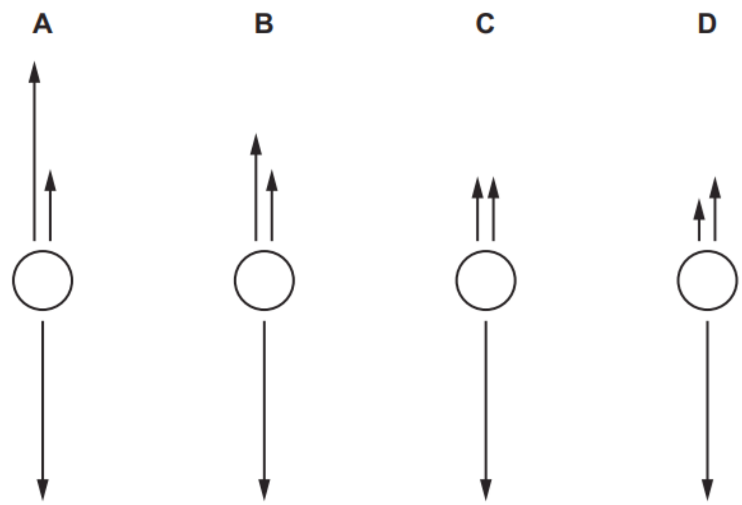
Which diagram shows the result of a perfectly elastic collision?



212. 9702/13/O/N/21/Q11

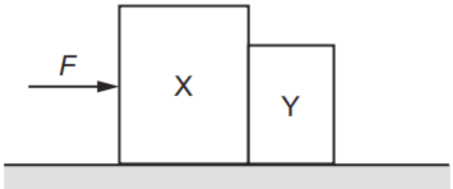
A spherical object falls through water at a constant speed. Three forces act on the object.

Which diagram, showing these three forces to scale, is correct?



213. 9702/12/F/M/22/Q7

A single horizontal force F is applied to a block X which is in contact with a separate block Y, as shown.



The blocks remain in contact as they accelerate along a horizontal frictionless surface. Air resistance is negligible. X has a greater mass than Y.

Which statement is correct?

A The acceleration of X is equal to force F divided by the mass of X.

B The force that X exerts on Y is equal to F.

C The force that X exerts on Y is less than F.

D The force that X exerts on Y is less than the force that Y exerts on X.

214. 9702/12/F/M/22/Q8

A car of mass 750 kg has a horizontal driving force of 2.0 kN acting on it. It has a forward horizontal acceleration of $2.0\,\mathrm{m\,s^{-2}}$.

What is the resistive force acting horizontally?

A 0.50 kN **B** 1.5 kN **C** 2.0 kN **D** 3.5 kN

215. 9702/12/F/M/22/Q10

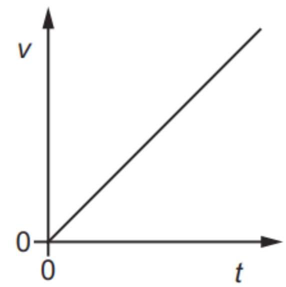
A rock of mass $2m$, travelling in deep space at velocity v, explodes into two parts of equal mass, one of which is then stationary.

What is the kinetic energy of the moving part after the explosion?

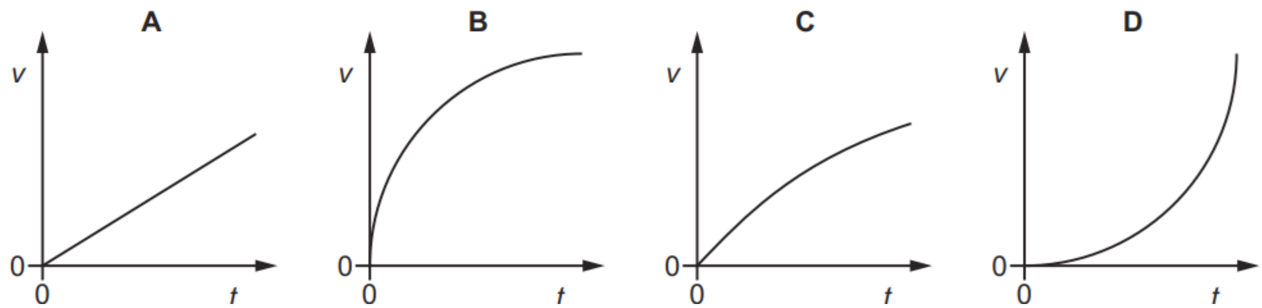
A $\frac{1}{2}mv^2$ **B** mv^2 **C** $\frac{3}{2}mv^2$ **D** $2mv^2$

216. 9702/12/F/M/22/Q9

An object falls freely from rest in a vacuum. The graph shows the variation with time t of the velocity v of the object.



Which graph, **using the same scales**, represents the object falling in air?



217. 9702/11/M/J/22/Q7

An object is moving along the ground in a straight line at a constant speed.

Which statement about the resultant force acting on the object is correct?

A The resultant force acting on the object is equal to its weight.

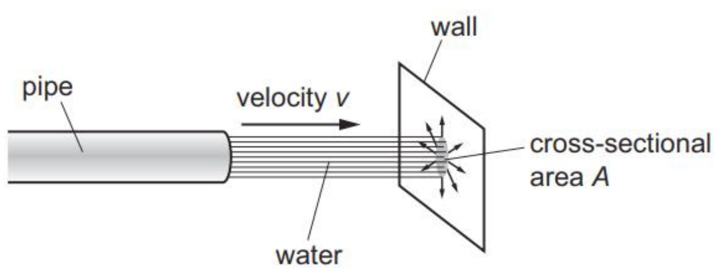
B The resultant force acting on the object is equal to the product of its mass and its velocity.

C The resultant force acting on the object is equal to the resistive force.

D The resultant force acting on the object is equal to zero.

218. 9702/11/M/J/22/Q8

Water flows out of a pipe and hits a wall.



When the jet of water hits the wall, it has horizontal velocity v and cross-sectional area A.

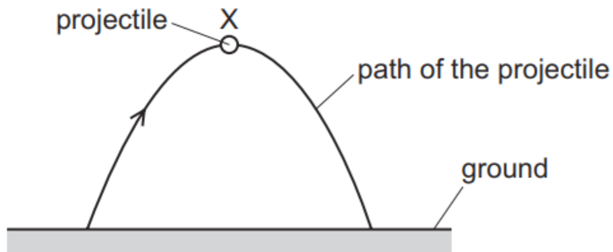
The density of the water is ρ. The water does not rebound from the wall.

What is the force exerted on the wall by the water?

A $\dfrac{\rho v}{A}$ **B** $\dfrac{\rho v^2}{A}$ **C** $\rho A v$ **D** $\rho A v^2$

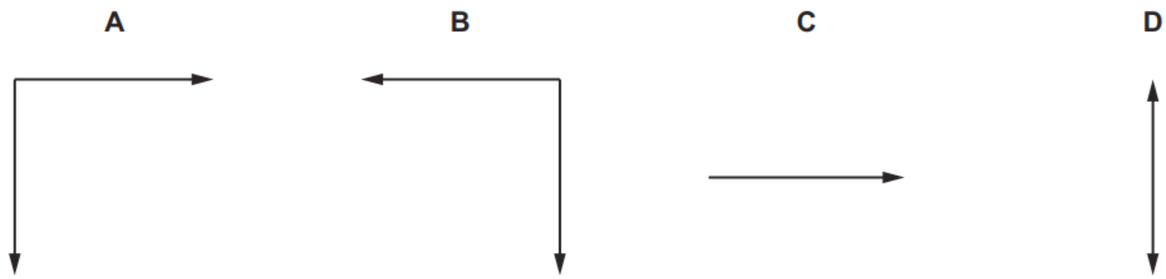
219. 9702/11/M/J/22/Q9

A projectile is launched at an angle above horizontal ground and travels through the air.



The projectile reaches its maximum height at position X. Assume that no upthrust acts on the projectile.

Which diagram shows the **directions** of the force or forces acting on the projectile at position X?



220. 9702/12/M/J/22/Q7

Which statement is **not** a requirement of a pair of forces that obey Newton's third law of motion?

A The forces act in opposite directions.

B The forces act on different objects.

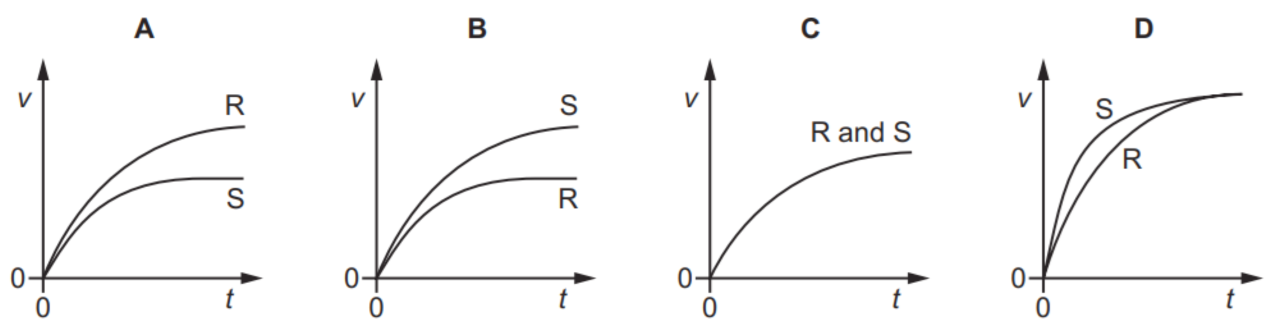
C The forces act on objects in contact.

D The forces are of equal magnitude.

221. 9702/12/M/J/22/Q9

A stone S and a foam rubber ball R are identical spheres of equal volume. They are released from rest at time $t = 0$ and fall vertically through the air. Both reach terminal velocity.

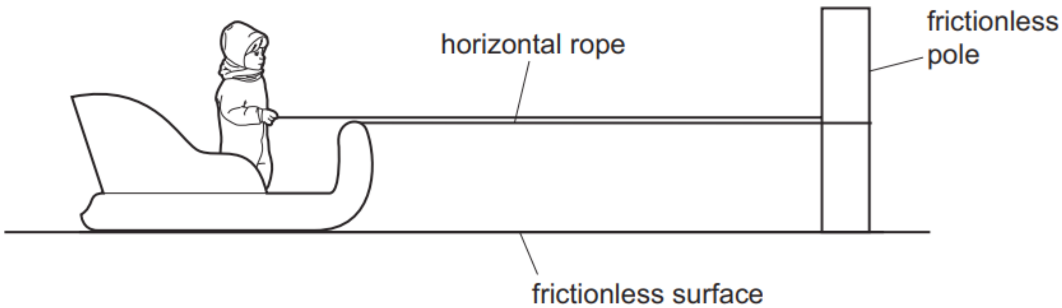
Which graph best shows the variation with time t of the speed v of the stone and of the rubber ball?



222. 9702/12/M/J/22/Q8

A child of mass 20 kg stands on the rough surface of a sledge of mass 40 kg. The sledge can slide on a horizontal frictionless surface.

One end of a rope is attached to the sledge. The rope passes around a fixed frictionless pole, and the other end of the rope is held by the child, as shown.



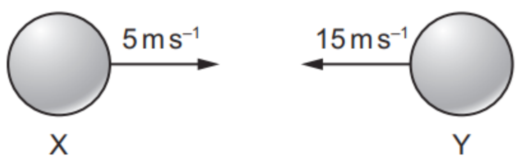
The rope is horizontal. The child pulls on the rope with a horizontal force of 12 N. This causes the child and the sledge to move with equal acceleration towards the pole.

What is the frictional force between the child and the sledge?

A 4.0 N **B** 6.0 N **C** 8.0 N **D** 12 N

223. 9702/12/M/J/22/Q10

Two balls X and Y are moving towards each other with speeds of $5\,m\,s^{-1}$ and $15\,m\,s^{-1}$ respectively.



They make a perfectly elastic head-on collision and ball Y moves to the right with a speed of $7\,m\,s^{-1}$.

What is the speed and direction of ball X after the collision?

A $3\,m\,s^{-1}$ to the left

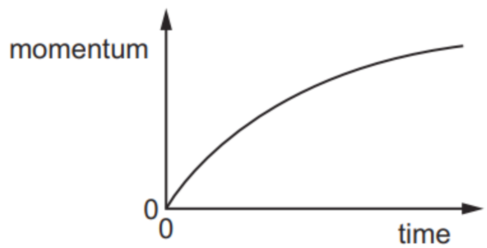
B $13\,m\,s^{-1}$ to the left

C $3\,m\,s^{-1}$ to the right

D $13\,m\,s^{-1}$ to the right

224. 9702/13/M/J/22/Q8

A car accelerates from rest. The graph shows the variation of the momentum of the car with time.



What is the meaning of the gradient of the graph at a particular time?

A the kinetic energy of the car

B the rate of change of kinetic energy of the car

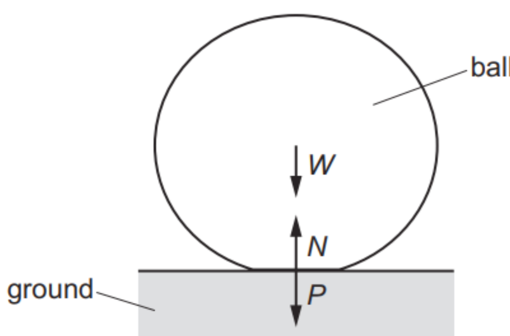
C the resultant force on the car

D the velocity of the car

225. 9702/13/M/J/22/Q9

A ball is dropped onto horizontal ground and bounces vertically upwards. When the ball is in contact with the ground, the following forces act:

- the weight W of the ball
- the contact force P exerted on the ground by the ball
- the contact force N exerted on the ball by the ground.



When the ball is in contact with the ground, the ball is momentarily stationary.

At this instant, which relationship is correct?

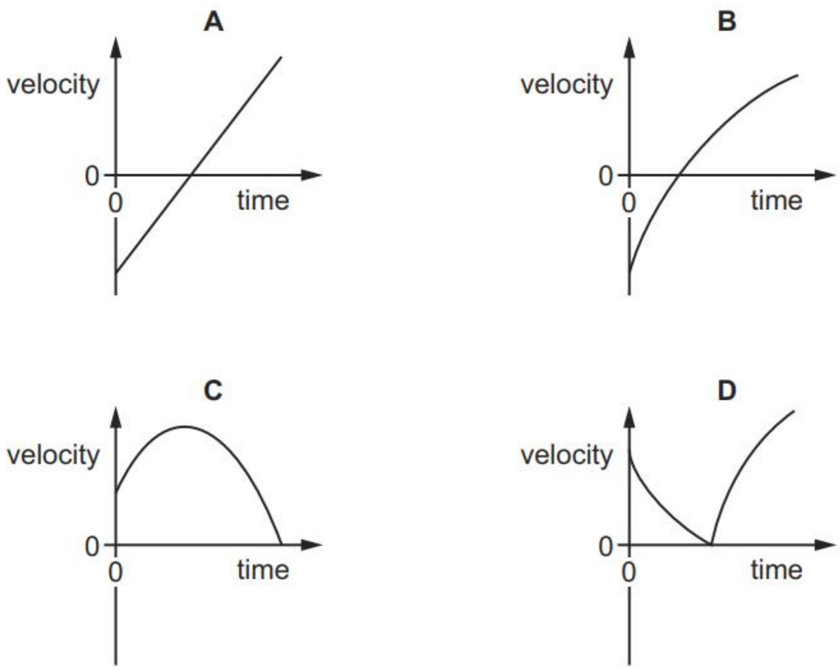
A $N = P + W$ **B** $N > P + W$ **C** $N = W$ **D** $N > W$

226. 9702/13/M/J/22/Q10

A person stands on the edge of a high cliff that is next to the sea. The person throws a stone vertically upwards. Air resistance acts on the stone.

The stone eventually hits the sea.

Which velocity–time graph best shows the motion of the stone from when it is released until it hits the sea?



227. 9702/13/M/J/22/Q11

Skaters of masses 80 kg and 40 kg move directly towards each other and collide.

Before the collision, the heavier skater is moving to the right at a speed of $2.0\,m\,s^{-1}$ and the lighter skater is moving to the left at a speed of $1.0\,m\,s^{-1}$.

After the collision, the heavier skater moves to the right at a speed of $0.80\,m\,s^{-1}$.

What is the relative speed of separation of the two skaters?

A $0.6\,m\,s^{-1}$ **B** $1.4\,m\,s^{-1}$ **C** $2.2\,m\,s^{-1}$ **D** $2.6\,m\,s^{-1}$

228. 9702/11/O/N/22/Q7

What is the definition of linear momentum?

A force per unit time

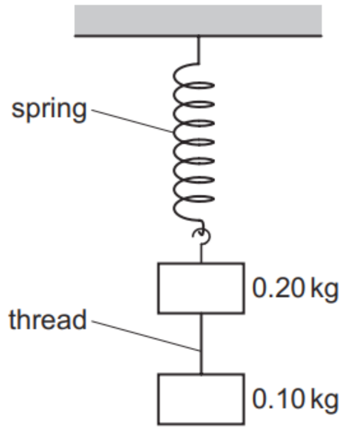
B product of force and time

C product of velocity and mass

D velocity per unit mass

229. 9702/11/O/N/22/Q8

A mass of 0.20 kg is suspended from the lower end of a light spring. A second mass of 0.10 kg is suspended from the first mass by a thread. The arrangement is allowed to come into static equilibrium and then the thread is cut.



Immediately after the thread is cut, what is the upward acceleration of the 0.20 kg mass?

A $4.9\,m\,s^{-2}$ **B** $6.5\,m\,s^{-2}$ **C** $9.8\,m\,s^{-2}$ **D** $15\,m\,s^{-2}$

230. 9702/11/O/N/22/Q9

A snowflake and a raindrop are in still air. They both fall from rest at the same time and from the same height, far above the ground.

The snowflake and raindrop contain the same mass of water. Assume that there is no evaporation or melting. Also assume that, for a given speed, the drag force acting on the snowflake is greater than the drag force acting on the raindrop.

Which statement about the snowflake and raindrop is correct?

A The raindrop takes more time than the snowflake to reach terminal velocity.

B The raindrop takes more time than the snowflake to reach the ground.

C They reach the same terminal velocity.

D They take the same amount of time to reach the ground.

231. 9702/12/O/N/22/Q8

A constant resultant force F acts on an object of mass m for time t.

What is the change in momentum of the object?

A $\frac{F}{t}$ **B** $\frac{Ft}{m}$ **C** Ft **D** $\frac{F}{mt}$

232. 9702/12/O/N/22/Q9

The acceleration of free fall on the surface of planet P is one-tenth of that on the surface of planet Q.

On the surface of P, an object has a mass of 1.0 kg and a weight of 1.0 N.

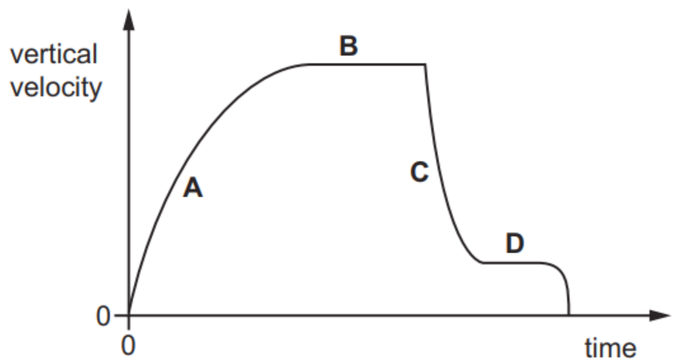
What are the mass and the weight of the same object on the surface of planet Q?

	mass on Q / kg	weight on Q / N
A	1.0	0.1
B	1.0	10
C	10	10
D	10	100

233. 9702/12/O/N/22/Q10

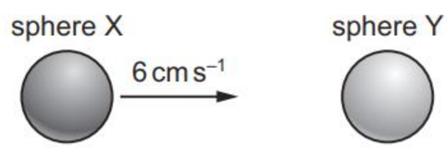
A parachutist falls from rest from a balloon. The variation with time of the vertical velocity of the parachutist is shown.

In which region is the force due to air resistance much greater than the weight of the parachutist?



234. 9702/12/O/N/22/Q11

Two solid spheres form an isolated system. Sphere X moves with speed $6\,\text{cm}\,\text{s}^{-1}$ in a straight line directly towards a stationary sphere Y, as shown.



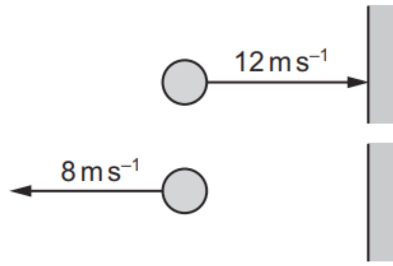
The spheres have a perfectly elastic collision. After the collision, sphere X moves with speed $2\,\text{cm}\,\text{s}^{-1}$ in the same direction as before the collision.

What is the speed of sphere Y?

A $2\,\text{cm}\,\text{s}^{-1}$ **B** $4\,\text{cm}\,\text{s}^{-1}$ **C** $6\,\text{cm}\,\text{s}^{-1}$ **D** $8\,\text{cm}\,\text{s}^{-1}$

235. 9702/13/O/N/22/Q8

A ball of mass 0.5 kg hits a vertical wall at a speed of $12\,m\,s^{-1}$. It bounces back along its original path with a speed of $8\,m\,s^{-1}$. The collision lasts for 0.10 s.

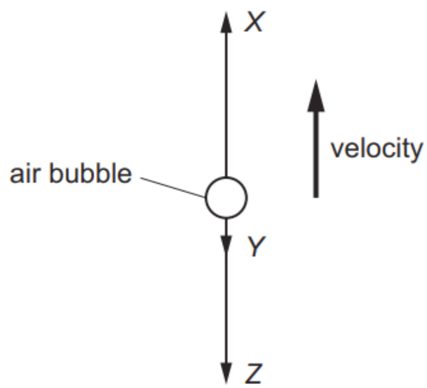


What is the average force on the ball due to the collision?

A 0.2 N **B** 1 N **C** 20 N **D** 100 N

236. 9702/13/O/N/22/Q9

An air bubble in a tank of water is rising with terminal (constant) velocity. The forces acting on the bubble are *X*, *Y* and *Z*, as shown.



The upthrust on the bubble is *X*.

Which statement about the forces is correct?

A *Z* is the viscous force on the bubble, *Y* is the weight of the bubble and $X = Y + Z$.

B *Z* is the viscous force on the bubble, *Y* is the weight of the bubble and $X > Y + Z$.

C *Z* is the weight of the bubble, *Y* is the viscous force on the bubble and $X = Y + Z$.

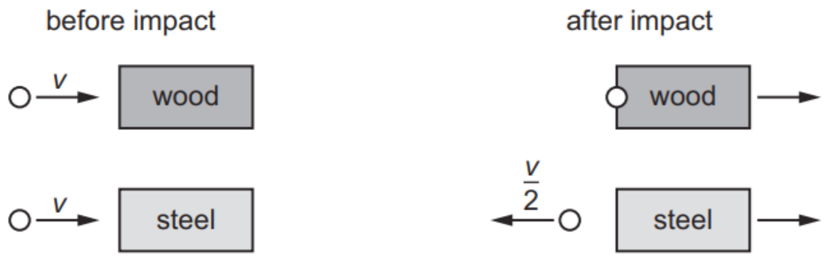
D *Z* is the weight of the bubble, *Y* is the viscous force on the bubble and $X > Y + Z$.

237. 9702/13/O/N/22/Q10

Two blocks are at rest on a frictionless horizontal surface. One block is made of wood and the other block is made of steel.

A steel ball is fired horizontally with a speed v at the wooden block. The ball embeds itself in the block, and the ball and block move together after impact.

A second identical steel ball is fired horizontally with speed v at the steel block. The steel ball then rebounds back along its original path with speed $\frac{v}{2}$.



The wooden block and the steel block have equal mass.

Which statement about the blocks immediately after the collisions is correct?

A Both blocks must travel with the same speed.

B The steel block must travel faster than the wooden block.

C The wooden block must travel faster than the steel block.

D The masses of the blocks and the steel ball are needed to determine which block travels faster.

238. 9702/12/F/M/23/Q7

Which expression defines force?

A (mass $\times$ change in speed) $\times$ time taken

B $\dfrac{\text{mass} \times \text{change in speed}}{\text{time taken}}$

C (change of momentum) $\times$ time taken

D $\dfrac{\text{change of momentum}}{\text{time taken}}$

239. 9702/12/F/M/23/Q8

A ship of mass 8.4×10^7 kg is approaching a harbour with speed $16.4\,\text{m}\,\text{s}^{-1}$. By using reverse thrust it can maintain a constant total stopping force of 920 000 N.

How long will it take to stop?

A 15 seconds

B 150 seconds

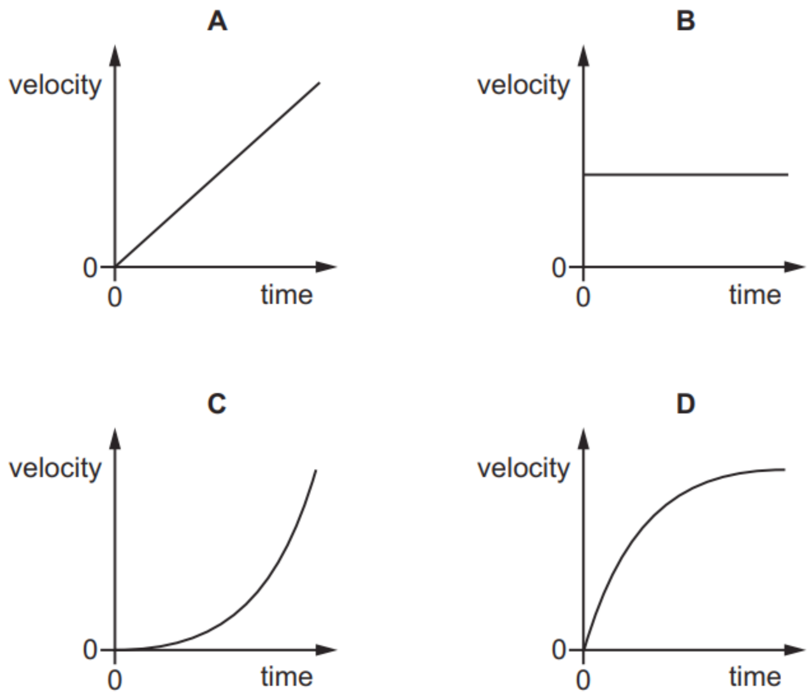
C 25 minutes

D 250 minutes

240. 9702/12/F/M/23/Q9

The velocity–time graphs of four different objects are shown.

Which graph represents an object falling from rest through a long distance in the Earth's atmosphere?



241. 9702/12/F/M/23/Q10

Which statement about collisions is correct?

A Kinetic energy is conserved in all collisions.

B Momentum is only conserved in perfectly elastic collisions.

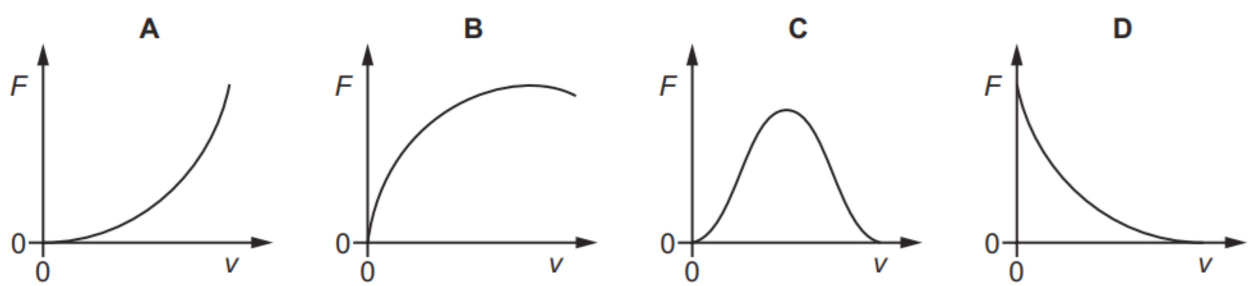
C The relative speed of approach is equal to the relative speed of separation for perfectly elastic collisions.

D When two objects of different masses collide, they exert forces of different magnitudes on each other.

242. 9702/11/M/J/23/Q9

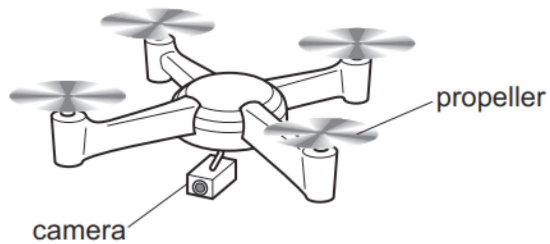
A small ball is held at the surface of liquid oil in a container. The ball is released from rest and falls through the oil. The ball has velocity v. A viscous (drag) force F acts on the ball.

Which graph could show the variation with v of F?



243. 9702/11/M/J/23/Q7

A camera drone of mass 1.20 kg hovers at a fixed point above the ground. The drone has four propellers.



In a time of 1.00 s, each propeller pushes a mass of 0.400 kg of air vertically downwards.

Assume that the air above the propellers is stationary.

What is the speed of the air leaving each propeller?

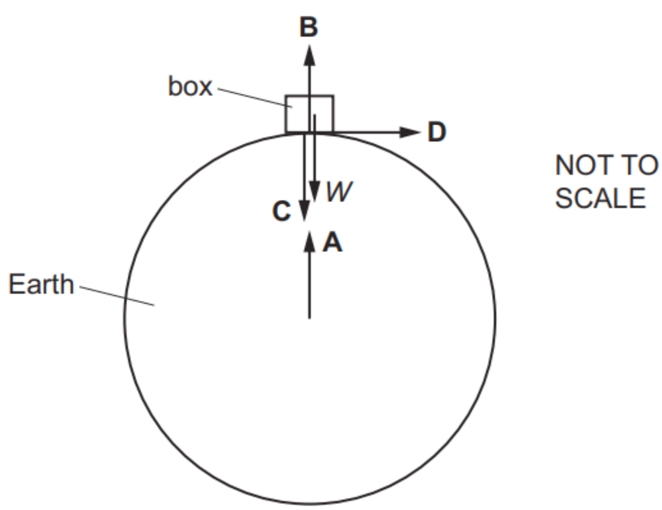
A $0.750\,\text{m}\,\text{s}^{-1}$ **B** $3.00\,\text{m}\,\text{s}^{-1}$ **C** $7.36\,\text{m}\,\text{s}^{-1}$ **D** $29.4\,\text{m}\,\text{s}^{-1}$

244. 9702/11/M/J/23/Q8

A box rests on the Earth, as shown.

Newton's third law describes how forces of the same type act in pairs. One of the forces of a pair is the weight *W* of the box.

Which arrow represents the other force of this pair?



245. 9702/11/M/J/23/Q10

An object of mass m, moving at speed u along a frictionless horizontal surface, collides head-on with a stationary object of mass $4m$.

before the collision

After the collision, the object of mass m rebounds along its initial path with $\frac{1}{4}$ of its kinetic energy before the collision.

What is the speed of the object of mass $4m$ after the collision?

A $\frac{u}{8}$ **B** $\frac{3u}{16}$ **C** $\frac{5u}{16}$ **D** $\frac{3u}{8}$

246. 9702/12/M/J/23/Q7

Which statement describes the mass of an object?

A the force the object experiences due to gravity

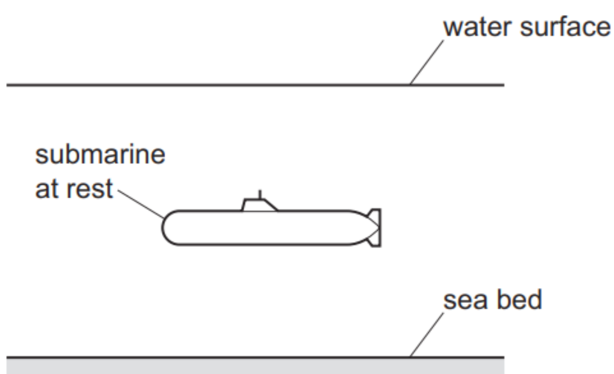
B the momentum of the object before a collision

C the resistance of the object to changes in motion

D the weight of the object as measured by a balance

247. 9702/12/M/J/23/Q8

A submarine of total mass 3200 kg is at rest underwater.



The total mass of the submarine is suddenly decreased by 200 kg by pumping water out of the submarine horizontally in a negligible time. The upthrust acting on the submarine is unchanged.

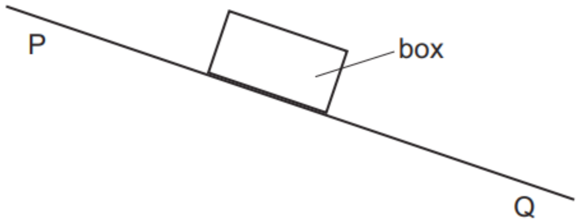
The change in the total weight of the submarine causes it to accelerate vertically upwards.

What is the initial upwards acceleration of the submarine?

A $0.613\,\text{m s}^{-2}$ **B** $0.654\,\text{m s}^{-2}$ **C** $9.81\,\text{m s}^{-2}$ **D** $10.5\,\text{m s}^{-2}$

248. 9702/12/M/J/23/Q9

A box in air slides with increasing speed down a rough slope from point P to point Q.



The slope surface exerts a constant frictional force on the box.

As the box moves from P to Q, there are changes to the magnitudes of its acceleration and the total resistive force acting on it.

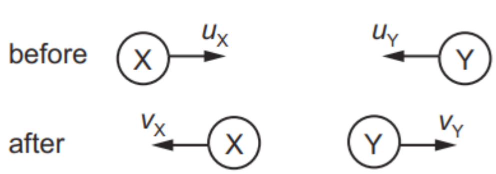
Which row describes the changes?

	magnitude of acceleration	magnitude of total resistive force
A	increases	decreases
B	decreases	decreases
C	increases	increases
D	decreases	increases

249. 9702/12/M/J/23/Q10

Two balls, X and Y, approach each other along the same straight line and collide. The collision is perfectly elastic.

Their initial speeds are u_X and u_Y respectively. After the collision they move apart with speeds v_X and v_Y respectively. Their directions are shown.



Which equation is correct?

A $u_X + u_Y = v_X + v_Y$

B $u_X + u_Y = v_X - v_Y$

C $u_X - u_Y = v_X + v_Y$

D $u_X - u_Y = v_X - v_Y$

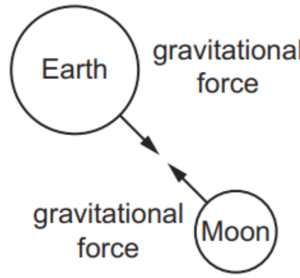
250. 9702/12/M/J/23/Q8

Each diagram illustrates a pair of forces of equal magnitude.

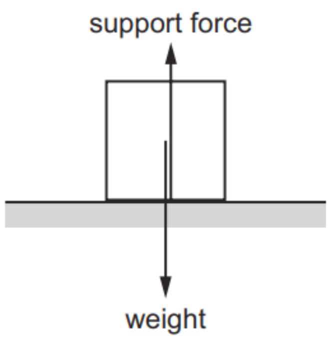
Which diagram gives an example of a pair of forces that is described by Newton's third law of motion?

A

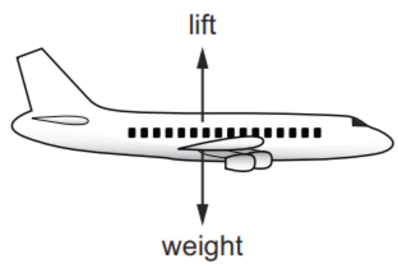
B



C



D



251. 9702/12/M/J/23/Q9

Two balls of identical shape and size but different masses are falling through the same liquid.

The sum of the drag force and upthrust acting on each ball is equal to its weight.

Which statement about the two balls is correct?

A The heavier ball has a larger acceleration than the lighter ball.

B The heavier ball has a smaller deceleration than the lighter ball.

C The heavier ball is falling at the same speed as the lighter ball.

D The heavier ball is falling at a larger speed than the lighter ball.

252. 9702/12/M/J/23/Q10

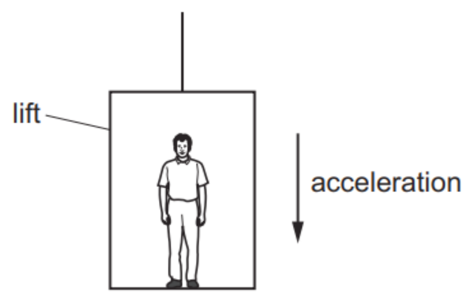
A perfectly elastic collision occurs between two objects X and Y. The mass of X is m and the mass of Y is $4m$. Object X travels at speed v before the collision and speed $\frac{3v}{5}$ in the opposite direction after the collision. Object Y is stationary before the collision.

What is the kinetic energy of Y after the collision?

A $\frac{8}{10}mv^2$ **B** $\frac{34}{50}mv^2$ **C** $\frac{16}{50}mv^2$ **D** $\frac{1}{5}mv^2$

253. 9702/11/O/N/23/Q7

A man stands in a lift that is accelerating vertically downwards, as shown.



Which statement describes the force exerted by the man on the floor?

A It is equal to the weight of the man.

B It is greater than the force exerted by the floor on the man.

C It is less than the force exerted by the floor on the man.

D It is less than the weight of the man.

254. 9702/11/O/N/23/Q8

A ball of mass 200 g is thrown horizontally with a speed of $20\,m\,s^{-1}$ against a vertical wall.

The ball is in contact with the wall for a time of 0.10 s before rebounding back along its original path with a speed of $10\,m\,s^{-1}$.

What is the average force exerted by the wall on the ball during the collision?

A 20 N **B** 60 N **C** 20 kN **D** 60 kN

255. 9702/11/O/N/23/Q9

In an experiment, a metal ball is dropped into a viscous liquid. The terminal velocity of the ball in the liquid is measured.

The experiment is repeated four times. For each repeat, a change is made to one of the following.

1 the density of the metal of the ball

2 the height from which the ball is dropped

3 the density of the liquid

4 the depth of the liquid

Which two changes separately affect the terminal velocity of the ball in the liquid?

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

256. 9702/11/O/N/23/Q10

Two objects move towards each other along the same straight line.

After colliding, the two objects stick together and are stationary.

Which statement **must** be correct?

A The total kinetic energy of the two objects does not change during the collision.

B The total momentum of the two objects before the collision is zero.

C The two objects have equal mass.

D The two objects have the same speed before the collision.

257. 9702/12/O/N/23/Q10

Two balls, of masses m and $2m$, travelling in a vacuum with initial velocities $2v$ and v respectively, collide with each other head-on, as shown.

After the collision, the ball of mass m rebounds to the left with velocity v.

What is the loss of kinetic energy in the collision?

A $\frac{3}{4}mv^2$ **B** $\frac{3}{2}mv^2$ **C** $\frac{9}{4}mv^2$ **D** $\frac{9}{2}mv^2$

258. 9702/12/O/N/23/Q9

An object falls from a stationary helicopter and reaches terminal velocity.

What happens to the acceleration of the object between leaving the helicopter and reaching terminal velocity?

A It decreases to $9.81\,\text{m}\,\text{s}^{-2}$.

B It decreases to zero.

C It increases to $9.81\,\text{m}\,\text{s}^{-2}$.

D It remains constant at $9.81\,\text{m}\,\text{s}^{-2}$.

259. 9702/12/O/N/23/Q8

A resultant force causes an object to accelerate.

What is equal to the resultant force?

A the acceleration of the object per unit mass

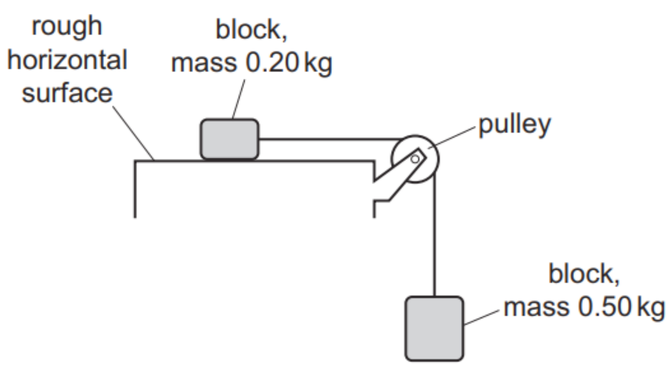
B the change in kinetic energy of the object per unit time

C the change in momentum of the object per unit time

D the change in velocity of the object per unit time

260. 9702/12/O/N/23/Q7

Two blocks, of mass 0.20 kg and 0.50 kg, are connected by a light inextensible string that passes over a frictionless pulley.



The blocks are initially held stationary. The block of mass 0.20 kg rests on a rough horizontal surface.

The block of mass 0.50 kg is suspended in air. Air resistance is negligible.

When the blocks are released, they have an acceleration of magnitude $2.0\,\text{m}\,\text{s}^{-2}$.

What is the magnitude of the frictional force between the block of mass 0.20 kg and the rough surface?

A 3.5 N **B** 3.9 N **C** 4.5 N **D** 6.3 N

261. 9702/13/O/N/23/Q7

Which statement about mass is correct?

A Mass has a magnitude and a direction.

B Mass resists changes in motion.

C The greater the mass of an object, the greater its acceleration when falling in a vacuum.

D The mass of an object depends on its location.

262. 9702/13/O/N/23/Q8

A snooker ball has a mass of 200 g. It hits the cushion of a snooker table and rebounds along its original path.

The ball arrives at the cushion with a speed of $14.0\,m\,s^{-1}$ and then leaves it with a speed of $7.0\,m\,s^{-1}$. The ball and the cushion are in contact for a time of 0.60 s.

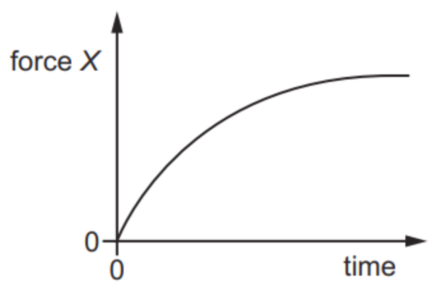
What is the average force exerted on the ball by the cushion?

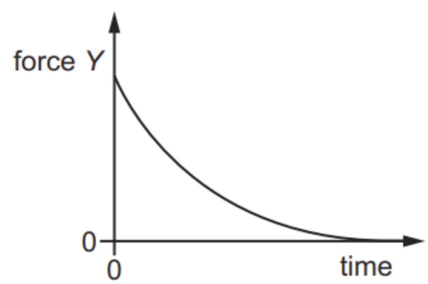
A 1.4 N **B** 2.3 N **C** 4.2 N **D** 7.0 N

263. 9702/13/O/N/23/Q9

A ball falls from rest through air and eventually reaches a constant velocity.

For this fall, forces *X* and *Y* vary with time as shown.





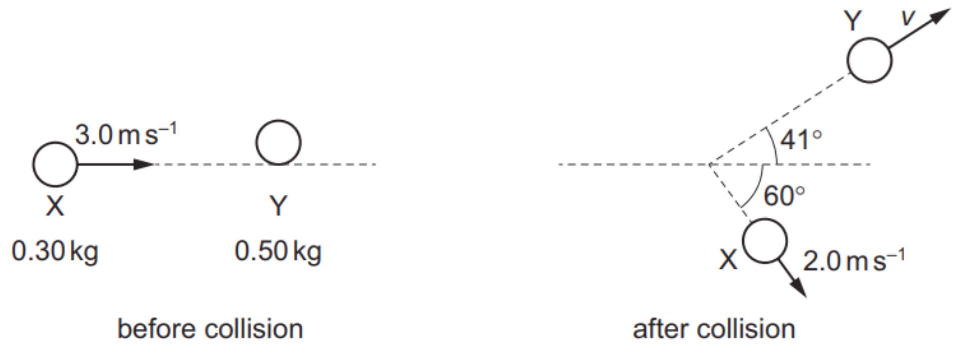
What could be forces *X* and *Y*?

	force *X*	force *Y*
A	air resistance	resultant force
B	air resistance	weight
C	upthrust	resultant force
D	upthrust	weight

264. 9702/13/O/N/23/Q10

An object X of mass 0.30 kg is travelling in a straight line at a constant velocity of 3.0 m s^{-1} on a horizontal frictionless surface. Object X collides with a stationary object Y of mass 0.50 kg.

After the collision, X moves with a velocity of 2.0 m s^{-1} at an angle of 60° to its direction before the collision. Object Y moves with a velocity *v* at an angle of 41° to the direction of X before the collision, as shown.



What is the value of *v*?

A 0.80 m s^{-1} **B** 1.2 m s^{-1} **C** 1.6 m s^{-1} **D** 1.8 m s^{-1}